J. Mol. Biol. (1992) 228, 220–242

Anatomy and Evolution of Proteins Displaying the Viral Capsid Jellyroll Topology

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(Received 25 February 1992; accepted 7 July 1992)

In this paper the anatomy of 25 structures containing a jellyroll motif, consisting of eight antiparallel β-strands forming a so-called β-barrel, was investigated. This involved performing a careful structural alignment based on hydrogen bonds for the equivalent regions of the tertiary folds and a subsequent analysis of conserved amino acids, equivalenced residue–residue contacts, and various parameters describing the size, shape and other geometrical characteristics of these regions. It was found that the jellyroll motif is best viewed as a two-sheet wedge structure rather than a barrel. The more conserved parameters are discussed. A model of evolutionary development for the jellyroll fold in the various protein and viral structures is proposed.

Keywords: protein folding; evolution; virus structure; protein structure; β-barrels

1. Introduction

The term "jellyroll" was first introduced by Richardson (1981) to describe a more complicated Greek key topology involving an extra swirl over other structures such as the immunoglobulin fold. The jellyroll motif was further discussed in the context of antiparallel β-strand domains that form roughly cylindrical or barrel shapes without necessarily maintaining a continuous hydrogen bonding pattern. Although these structures did not form as regular or symmetric a barrel as in the cases of the parallel stranded β-barrels of triosephosphate isomerase or pyruvate kinase, it was argued that a common barrel description yielded a simpler and more unified topological description.

The same jellyroll topology has been described by others as a wedge shape (Filman & Hogel, 1991; Stubbs, 1989), an RNA virus capsid domain (Harrison, 1991), a β-barrel (Abad-Zapatero *et al.*, 1980), a β-sandwich (Jones *et al.*, 1989), and an eight-stranded antiparallel β-barrel fold with a β-roll topology (Hosur, 1987). The reason for the diversity in nomenclature is that the structured architecture of the fold does, at least in parts, exhibit properties consistent with these naming schemes. Here, the term jellyroll will be adopted to describe these structures because its nature invokes a comprehensible mental picture of the somewhat complex topology. Further, it will be shown that this unique fold does not possess certain characteristics found in other β-barrel structures and thus should not be referred to as such.

The jellyroll motif is illustrated in Figure 1A. It may be considered as a single long hairpin coiled in a helical manner to form two four-stranded antiparallel β-sheets which pack together. For historical reasons, the strands of the jellyroll have been labeled B through I with strands B, I, D and G forming one sheet (the BIDG sheet) and strands C, H, E and F forming the other (the CHEF sheet). One side of the hairpin is composed of B, C, D and E strands while the other hairpin side is comprised of the F, G, H and I extended structures. The shorthand nomenclature for the sheets illustrates that strand B is hydrogen-bonded to strand I, I to B and D, D to I and G, and finally G to D. The same pattern holds for the CHEF sheet. While the major framework or topology of the jellyroll structures is the same, they differ drastically in terms of their interconnecting loops. Various embellishments ranging from extensive regions of coil, additional sheets or helices are permissible. The structurally conserved regions can show very poor sequence homology despite the complicated topology.

Each of the 25 structures used in this study

0022-2836/92/210220-23 $08.00/0

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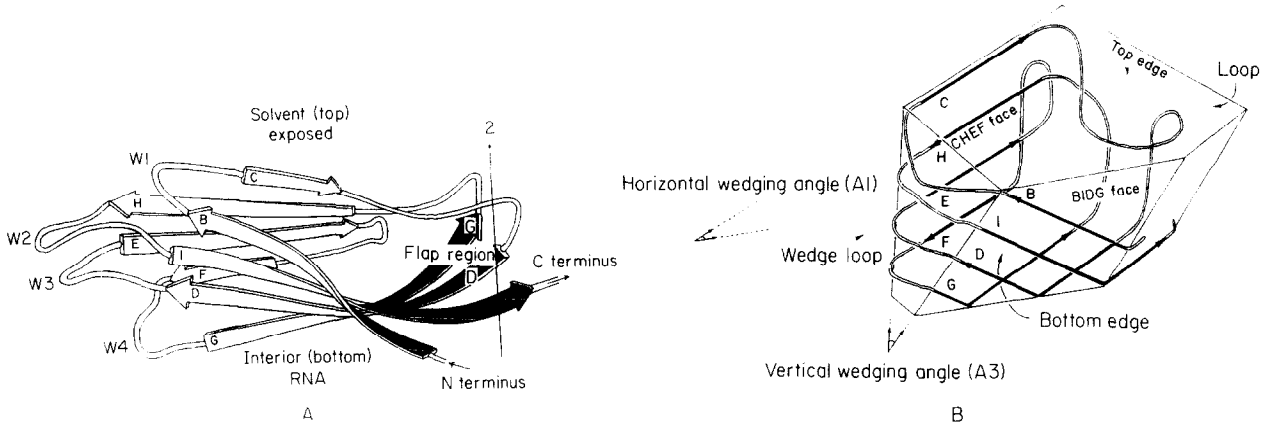

Figure 1. A. The jellyroll fold consisting of 8 β-strands (B through I) forming 2 antiparallel sheets. The wedge shape is apparent with its sharp end formed by loops W1 to W4 connecting strands B and C, I and H, D and E and F and G, respectively. The shaded region indicates the flap region. The fold is oriented relative to the indicated 2-fold axis for structures with a triangulation number of 3 (Caspar & Klug, 1962). B. The intrinsic wedge shape of the jellyroll resulting in 6 subunit interaction surfaces: the 2 side faces, the 2 loop ends and the top and bottom edges. The BIDG and CHEF sheets form the sides of the wedge. The planes fitted to their respective strands are designated as face interaction surfaces. The wedge loops, W1 to W4, as well as the longer CD, EF and GH loops form the ends of the wedge. These ends are designated as loop interaction surfaces. The top of the wedge involves the B and C strands in addition to parts of the surface loops while the bottom is bounded by the F and G strands and includes parts of the flap. The latter interaction surfaces are designated as edge. Indicated also are the vertical and horizontal wedging angles used to describe the orientation of the BIDG and CHEF faces with respect to one another.

contains a characteristic jellyroll motif. Table 1 lists them along with their identification code and names (where applicable) in the tertiary structural database (Bernstein *et al.*, 1977), icosahedral symmetry (in the case of virus capsid structures), resolution and appropriate references reporting the three-dimensional folds. The structures can be categorized either on the basis of their genetic encoding or their icosahedral symmetry. Seven groups are distinguished here: picornaviruses, comoviruses, $T = 3$ plant viruses, $T = 1$ plant viruses, nodaviruses, human DNA viruses and non-virus structures. The members of each of these classes are described briefly below.

Picornaviruses, which include rhino (RHV† and R1A), mengo (MEV), and polio (PLV) viruses, are so-called because their virions are among the smallest RNA-containing animal viruses. These viruses utilize an icosahedral capsid of 60 copies of each of four subunits labeled VP1 to VP4 with an external diameter of approximately 300 Å (1 Å = 0·1 nm). Three of the subunits, VP1 to VP3, are of roughly similar size and each displays a jellyroll topology. VP4 is comparatively small (~70 residues), occurs internal to the mature virion and is thought to be involved with the final stages of assembly as well as in the initial steps of the viral uncoating (Koch & Koch, 1985). The outer surface of the capsid is formed by VP1, VP2 and VP3, which arrange themselves to form a pseudo $T = 3$ icosahedron ($P = 3$) according to the nomenclature of Caspar & Klug (1962). This triangulation is referred to as pseudo since the asymmetric unit is composed of three different proteins, albeit with similar fold and related by an approximate or quasi-3-fold axis (Rossmann & Johnson, 1989). Normally a $T = 3$ virus would use 180 copies of the same protein.

Members of the comovirus group include bean-pod mottle virus (BMV), cowpea mosaic virus (CPMV) and red clover mottle virus (RCMV) (Lomonossoff & Johnson, 1991) although only BMV, with known tertiary structure, is considered here. Picornaviruses share a number of physical and biological properties with comoviruses (Rossmann & Johnson, 1989). While virions of both virus types show $P = 3$ icosahedral shells of similar diameter, the capsid of comoviruses is formed from only two promoters, designated S for the small and L for the large subunits. The L subunit is composed of two domains (B and C) and by comparing the genes and the relative subunit positioning in the capsids in picornaviruses and comoviruses (Chen *et al.*, 1989), it is seen that the S subunit (denoted the A chain) corresponds directly to VP1 while the B domain corresponds to VP3 and the C domain to VP2.

The RNA plant viruses include tomato bushy stunt virus (TBV), southern bean mosaic virus (SBV) and satellite tobacco necrosis virus (STV). TBV and SBV form $T = 3$ icosahedral capsids and are composed of 180 covalently identical subunits.

† Abbreviations used: RHV, R1A, rhinoviruses; MEV, mengovirus; PLV, polio virus; BMV, bean-pod mottle virus; CPMV, cowpea mosaic virus; RCMV, red clover mottle virus; TBV, tomato bushy stunt virus; SBV, southern bean mosaic virus; STV, satellite tobacco necrosis virus; BBV, black beetle virus; HMG, hemaglutinin glycoprotein; TNF, tumor necrosis factor; GAP, gene activator protein; r.m.s., root-mean-square; FMDV, foot-and-mouth disease virus; CPV, canine parvovirus; FIX, bacteriophage ϕX174; CNA, concanavalin A.

Table 1
Codes and references for the jellyroll structures

Code	Virus	Symmetry	Resolution (Å)	Reference
MEV-1 MEV-2 MEV-3	Mengo encephalomyocarditis virus	Icosahedral $P = 3$	3·0	Krishnaswamy & Rossmann (1990)
RHV-1 RHV-2 RHV-3	Human rhinovirus	Icosahedral $P = 3$	3·0	Arnold & Rossmann (1988)
RIA-1 RIA-2 RIA-3	Rhinovirus serotype 1	Icosahedral $P = 3$	3·2	Kim *et al.* (1989)
PLV-1 PLV-2 PLV-3	Poliovirus	Icosahedral $P = 3$	2·9	Hogel *et al.* (1985)
BMV-A BMV-B BMV-C	Bean-pod mottle virus	Icosahedral $P = 3$	3·0	Chen *et al.* (1989)
SBV	Southern bean mosaic virus	Icosahedral $T = 3$	2·8	Silva & Rossmann (1985)
TBV	Tomato bushy stunt virus	Icosahedral $T = 3$	2·9	Harrison *et al.* (1978)
STV	Satellite tobacco necrosis virus	Icosahedral $T = 1$	3·0	Liljas *et al.* (1982)
BBV	Black beetle virus	Icosahedral $T = 3$	3·0	Hosur *et al.* (1987)
AHV-1 AHV-2	Adenovirus hexon	Trimeric	2·9	Roberts *et al.* (1986)
HMG	Influenza virus hemagglutinin	Trimeric	3·0	Weis *et al.* (1990)
TNF	Tumor necrosis factor	Trimeric	2·6	Eck & Sprang (1989)
GAP	Catabolite gene activator protein	Dimeric	2·5	Weber & Steitz (1987)
CNA	Concanavalin A	Dimeric	2·0	Reek *et al.* (1975)

TBV also contains another major domain called the P or protruding domain. While the S or surface domain displays a standard class I jellyroll motif (see Fig. 1A), the P domain, which projects out from the capsid, is the only known example of a class II topology (Gibson & Argos, 1990) and is thus not considered here. The arrangement of SBV subunits which contain no P domain is similar to the S domain of TBV. STV is a satellite virus which cannot replicate in the absence of tobacco necrosis virus. It forms $T = 1$ icosahedral capsids with a mean diameter of 170 Å requiring only 60 identical copies of a single subunit. While the topology of the STV subunit shows a clear similarity to those of TBV and SBV, the packing between subunits is markedly different around the 3 and 2-fold axes (Rossmann *et al.*, 1983; Johnson & Argos, 1985; Liljas, 1986).

Nodaviruses are small icosahedral viruses which infect insects as well as mammalian hosts. The only tertiary structure determined for this group is that of the black beetle virus (BBV) capsid. BBV utilizes 180 copies of a single capsid protein to form a $T = 3$ icosahedral coat with the arrangement of the protomer the same as in the $T = 3$ plant viruses.

Adenoviruses are non-enveloped DNA viruses that can infect mammalian hosts. They form large icosahedrally symmetric shells which involve a number of different constituent proteins. A total of 252 capsomers are visible through electron microscopy, 12 of which are pentamers and occur at icosahedral 5-fold axes, the remainder forming hexomer structures (Roberts *et al.*, 1986). The hexon is a trimer containing three identical polypeptide chains, each with three domains: P1, P2 and T. The P1 (AHV-1) and P2 (AHV-2) domains have a jellyroll topology which form the base portion of the trimer. Some of the loops connecting the jellyroll strands are large and help form the tower domain T.

The hemaglutinin glycoprotein of the influenza virus forms (HMG) trimers in the lipid membrane acquired by the maturing virion as it buds through the host-cell membrane. HMG is one of two virally coded integral membrane proteins, the other being neuraminidase (Wilson *et al.*, 1981). Each of the HMG trimer subunits consists of three domains: a five-stranded β-sheet at the membrane end of the molecule, a long helical domain which forms a coiled-coil with the equivalent domains in the other trimer subunits and a globular domain which contains the host-receptor binding site. The globular domain adopts a jellyroll conformation.

The group of structures considered as non-viral are tumor necrosis factor (TNF), concanavalin A (CNA) and the large domain of the catabolite gene activator protein (GAP). TNF belongs to a class of hormones termed cytokines which regulate the growth and differentiation of cells involved in inflammation, immunity and hemopoiesis (Jones *et al.*, 1989). The active form of TNF is believed to be trimeric involving an unusual edge-to-face packing of three covalently identical jellyroll topologies. Concanavalin A is a dimer of identical protomers and binds to a variety of eukaryotic cells through specific interactions with saccharide-containing cellular receptors (Reeke *et al.*, 1975; Becker *et al.*, 1975). The GAP protein regulates transcription of several catabolite-sensitive operons in *Escherichia coli* (Zubey *et al.*, 1970). Cyclic AMP acts as an allosteric activator and in its presence GAP binds to

specific DNA sequences near the protomer region of operons such as those for lactose. In its active form GAP is a dimer involving two identical subunits. Each subunit contains a large domain consisting of a jellyroll motif with a long helix serving to seal off the GF end of the jellyroll in addition to connecting the large domain to a smaller C-terminal helical domain. Cyclic AMP binds to the short loop interconnecting strands G and H and becomes buried within the interior of the large domain when the long helix packs against the jellyroll.

While this work was being conducted, several other structures containing jellyroll motifs were solved by X-ray crystallographic techniques; namely, the capsids of simian virus (SV40: Liddington *et al.*, 1991), parvovirus (Tsao *et al.*, 1991) and the bacteriophage ΦX174 (McKenna *et al.*, 1992). Also not used in this work is the tertiary structure of phaseolin, a seed storage protein (Lawrence *et al.*, 1990), for which only C^{α} co-ordinates have been deposited in the tertiary structure database.

2. Methods

(a) *Multiple structural alignment*

The results of a 1-dimensional sequence alignment can depend heavily on the method used and further on the values of threshold parameters chosen (e.g. gap penalties and similarity scores (for a review, see Argos *et al.*, 1991). A similar dependence on methodology is found for the results of 3-dimensional structural alignments (Taylor & Orengo, 1989; Sali & Blundell, 1990). Varying techniques exist to superimpose protein structures and the reader is referred to Matthews & Rossmann (1985) for a review. The procedures adopted here make use of the main-chain hydrogen bonding patterns found in antiparallel β-sheets. The effectiveness of superpositions based on hydrogen bonds for β-proteins has been previously discussed (Rossmann *et al.*, 1975; Sali & Blundell, 1990). Once such patterns are generated, the superpositions are effectively performed in 2-dimensions by sliding the patterns over one another to achieve the maximum number of superimposed hydrogen bonds.

The major innovation introduced here to determine the hydrogen bonding schemes was the relaxation of typical constraints to achieve as complete a pattern as possible. To generate a pattern, a list of main-chain hydrogen bonds was first derived allowing a maximal distance between an amide nitrogen and carbonyl oxygen (N-H . . . O) of 3·5 Å and an angular deviation of 60° and 180° in the N-H . . . O and C-O directions. From this list, bridges, ladders and partial sheets were constructed following from the definitions of Kabsch & Sander (1983). The incomplete pattern was then searched for positions where other hydrogen bonds might exist under slightly more relaxed constraints such as at the ends of ladders or in between 2 residues in antiparallel strands where only one of the 2 possible hydrogen bonds was found (see Fig. 2 for an example). The less stringent parameters included a maximal threshold for the N-H . . . O distance of 4·5 Å and the largest deviation angle allowed set at 70°. This method is not equivalent to simply starting bridge and ladder construction from a less constrained set of hydrogen bonds where relaxed constraints may allow bifurcated bonds to propagate leading to inconsistencies in bridge and ladder building.

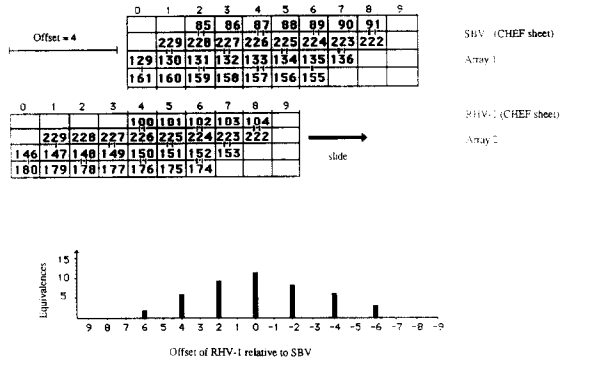

Figure 2. Illustration of the superposition process of the 2 hydrogen bond patterns for the CHEF sheets of SBV and RHV-1. Hydrogen bonds are indicated by short vertical bars between the residues they bond. A thicker bar is printed where such bonds were introduced as a result of the constraint relaxing process used during pattern construction. For superposition, each pattern is first stored in an array. One array is then slid over the other, testing at each register the number of positions with corresponding hydrogen bonds. Sliding starts with column 9 of array 2 immediately under column 0 of array 1 (corresponding to an offset of 9 which is the distance from column 0 of array 2 to column 0 of array 1). Sliding proceeds until column 0 of array 2 is under column 9 of array 1 (corresponding to an offset of −9). A score is calculated for each different offset position by summing the number of hydrogen-bonded residue pairs in array 1 that have analogs in array 2. For example, for an offset of 4 there are 6 such hydrogen-bonded pairs: residues 129–161, 229–130, 85–228, 131–159, 227–132 and 87–226 in array 1 with respective analogs 150–176, 225–151, 102–224, 152–174, 223–153 and 104–222 in array 2. The analogous residues for the offset with the highest score are taken as equivalent and a corresponding transformation matrix is calculated to superimpose them.

Hydrogen bond patterns found in β-structures can be projected into 2-dimensions if bulges are pruned. Placing 2 such patterns in 2-dimensional arrays provides the possibility of sliding one array over the other to attain the most superimposed hydrogen bonds (see Fig. 2 for an illustration). The residues corresponding to the superimposed hydrogen bonds can be assigned as equivalent and a transformation matrix to place one structure onto the other generated using the routine of Kabsch (1976). In the case where more than one relative position of the arrays in the previous step yields the same number of overlapping bonds, the superposition giving the lowest root-mean-square (r.m.s.) deviation value between the equivalenced C^{α} atoms was adopted.

For each of the jellyroll structures given in Table 1, hydrogen bond patterns were generated separately for the BIDG and CHEF sheets and these sheets were subsequently treated individually. All possible pairwise superpositions of equivalent sheets were performed. Using the equivalent C^{α} atom r.m.s. value as an indicator of similarity, a sorted list of relatedness between pairs of structures was built and traversed in descending order. Taking the closest pair as a start point, their equivalent residues were used to form a sequence alignment. Sequences were then added to old groups or new groups formed in

Figure 3. Superposition of all 25 jellyroll structures. Separate superpositions are given for the BIDG and CHEF sheets. Virtual bonds are shown between successive C^{α} atoms in the stick representations of the β-strands.

descending down the sorted list. Groups would be merged if each structure in the current pair taken from the list already existed in separate groups. Eventually one final and consistent multiple alignment for each of the BIDG and CHEF sheets resulted. At each step, the most consistent offset, derived from the relative positioning of the arrays, was taken to determine the equivalenced residues in the growing alignment. The separate superpositions for the BIDG and CHEF sheets for the structures given in Table 1 are illustrated in Fig. 3.

(b) *Linear and planar geometric constraints*

Linear and planar geometric parameters for simplified models of the jellyrolls in Table 1 were determined as outlined below. The parameters measured focus on the positioning and orientation of the strands within each of the sheets and of the sheets (BIDG and CHEF) with respect to one another. In particular, estimates were made for the intra-sheet strands, the inter-sheet strands, the inter-sheet distance and orientation and also the volume enclosed by the 2 sheets.

The BIDG sheet found in virus capsid proteins is easily recognized by its length and severe twist (Fig. 1A) which renders the use of a single plane a poor model. Consequently, by inspection of shifts in the hydrogen bond patterns, a cut was introduced so as to divide the sheet into 2 regions: one packing face-to-face with the CHEF sheet and the other interacting more with the CD, GH and EF loops. For the purpose of this study, the former region will be referred to as the BIDG sheet and the latter as the flap region. While the BIDG sheet is represented in all the jellyroll structures, the flap region varies greatly and in some instances is not present as in BMV-C. Fig. 7 indicates the β-strand flap region residues that could be structurally equivalenced amongst the various structures.

To effect the measurements, planes were fitted to the BIDG and CHEF sheets as defined in Fig. 4 by applying principle component analysis (Press *et al.*, 1988) to their

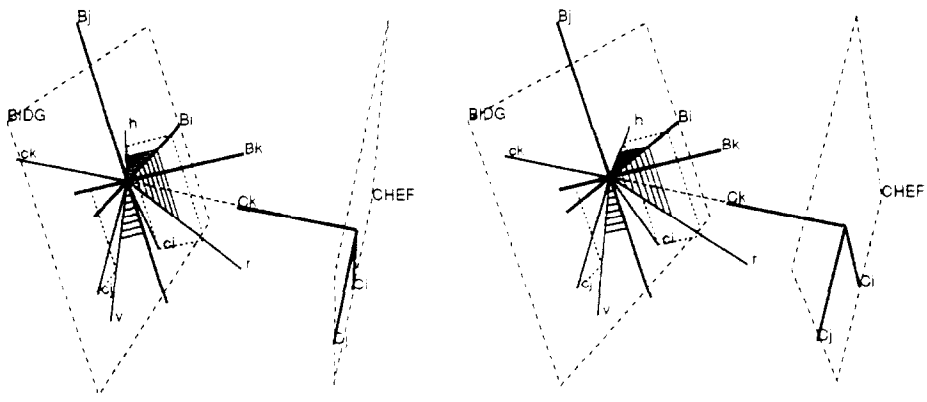

Figure 4. Definitions of the inter-plane geometric parameters. The planes fitted to the BIDG and CHEF sheets of MEV-2 are shown. The axis system of the CHEF sheet (C_i, C_j, C_k) is translated onto the axis system of the BIDG sheet (B_i, B_j, B_k) to form a new axis system (c_i, c_j, c_k). Both B_i and C_i have the same directional sense as the β-strand. The angle between the projection of c_i onto the B_iB_k plane (**h**) and the B_i axis is defined as π minus the horizontal wedging between the 2 sheet planes (A1 in Table 4). Likewise, the angle between the projection of c_i onto the B_iB_j plane (**r**) and the B_i axis is defined as the rotation between sheet planes (A2 in Table 4). The angle between the projection of c_j onto the B_jB_k plane (**v**) and the $-B_j$ axis is defined as the vertical wedging between the 2 planes (A3 in Table 4). The inter-plane distance (D in Table 4) is indicated by the broken line between the centers of mass of the 2 planes.

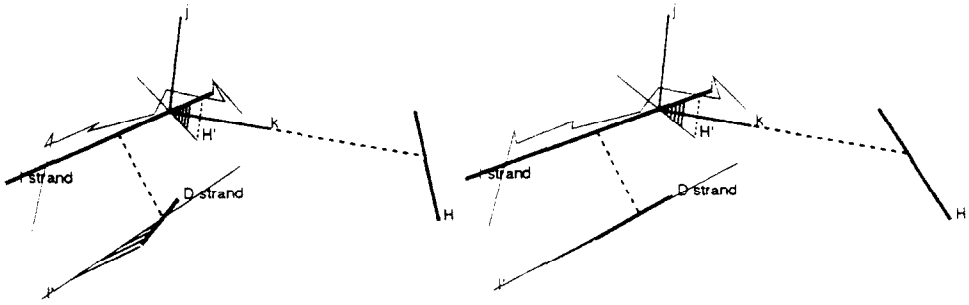

Figure 5. Definitions of inter- and intra-strand geometric parameters. A C^{α} trace of the I strand of MEV-2 is shown together with the line fitted as its axis. The axis lines for the D and H strands are shown also, connected to the I strand by closest approach lines (broken). A conventional right-handed axis system can be defined using a strand axis and the direction of the line of closest approach to another strand to indicate the **i** and **k** unit vectors, respectively. The rotations about the unit **k** axis necessary to bring the second strand axis into the i, k plane is defined as the vertical wedging when the strands are on opposite sheets or the inter-strand twist when the strands are on the same sheet. The rotation about the unit **j** axis required to bring the second axis into the i, j plane is defined as the horizontal wedging between structures on opposite sheets. In the example shown, this value is zero.

respective constituent C^{α} atomic co-ordinates. The 2 major axes determined a plane face while the third axis its normal. The same technique was applied to the C^{α} atoms of each strand, the major axis determining the direction of the line fitted to the strand. Using these simplifications, the intra-sheet strand distance was defined to be the magnitude of the vector, **v**, describing the distance of closest approach between consecutive strands in the sheet. The consecutive twist down the sheet was measured as the rotation angle, about **v**, required to align the strands (Fig. 5). The distance between the BIDG and the CHEF planes was taken as the distance between their respective centers-of-mass defined by the C^{α} atom co-ordinates. Three angles were used to describe the orientation of one plane with respect to another: A1, A2, A3. The angle A1 corresponds to the wedging between the sheets in the direction of their strands (horizontal wedging), while A2 corresponds to the rotation of one sheet with respect to the other (twist) and A3 corresponds to the wedging between the sheets in the direction perpendicular to the strands (vertical wedging). Illustrations for the definition of these angles are given in Fig. 4.

A similar approach was taken to obtain the inter-sheet strand measurements. The distance between 2 strands was taken as the displacement of their centers-of-mass. The horizontal wedging of one strand with respect to its counterpart in the opposite sheet (B/C, I/H, E/D and G/F) was given by A4 and was calculated in the same manner as A1. Likewise, A5, the rotation of one strand with respect to another was calculated as A2 above. Inter-sheet values are only given between equivalent strands in the 2 sheets since it was found that, on average, these strand pairs showed the closest proximity to one another.

The volume between the BIDG and CHEF sheets was estimated by summing the individual volumes of all the atoms in those residues considered to be interior to the jellyroll. Internal residues were defined as having an incident angle between 2 vectors **p** and **q** less than $\pi/2$. Here, **q** is defined as the vector from a residue's C^{α} atom to its side-chain center-of-mass. The vector **p** is perpendicular to the direction of the strand to which the current residue belongs and lies in the plane defined by the strand axis and line of closest approach to the opposite strand.

(c) *Twisted hyperboloid parameterization*

Proteins containing β-barrel motifs have been modeled by cylinders (McLachlan, 1979) and later as twisted hyperboloids (strophoids) by Novotny *et al.* (1984). Following the latter work, Lasters *et al.* (1988) developed a simplified method to approximate hyperboloid parameters. A similar technique was adopted here to derive such parameters and is outlined as follows. Lines were fitted to each of the β-strands, excluding the flap region, using the procedure described earlier. The averaged directional cosines of the strands was taken as the hyperboloid axis. The waist of the hyperboloid was determined by iteratively sliding a plane, using the hyperboloid axis as a normal, down the hyperboloid axis to locate the minimal sum of the distances between the intersection point of each of the strand lines with the plane and the intersection point, o, of the axis with the plane. The major and minor semi-axes, denoted A and B, respectively, were determined by fitting a least-squares ellipse, with center o, to the intersection points of the strand lines with the waist plane. The helical pitch of the twisted hyperboloid was estimated by fitting further ellipses other than the one at the waist at 1 Å intervals along the hyperboloid axis and averaging the rotation angles (T) between their consecutive major semi-axes. The area (Ar) of the waist ellipse was calculated as $Ar = AB\pi$, as well as the ratio R of the magnitudes of the semi-axes, given by $|A|/|B|$ and then the eccentricity, E, of the waist ellipse. An exemplary hyperboloid fit is given in Fig. 6.

(d) *Residue–residue contacts*

Intra-jellyroll residue–residue contacts were generated for each of the 25 structures listed in Table 1. Any 2 residues were taken to be in contact if the distance between the centers of any pair of atoms, one contributed from each residue, was less than the sum of their respective van der Waals radii, as defined by Stryer (1981), plus a constant distance of 1·5 Å to account for attached hydrogen atoms and errors in the X-ray crystallographically determined structures.

In order to investigate the possibility of specific residue–residue interactions providing a global mechanism for the stability or folding of the jellyroll unit, contact maps were generated for each of the sequences aligned in Fig. 7. A contact map for one sequence is a symmetric matrix with the strand alignment sites corresponding to the matrix row and column elements. Binary values are entered in each of the possible residue–residue contact cells if a contacting atom pair is found (1) or not found (0). As each column in the alignment corresponds to

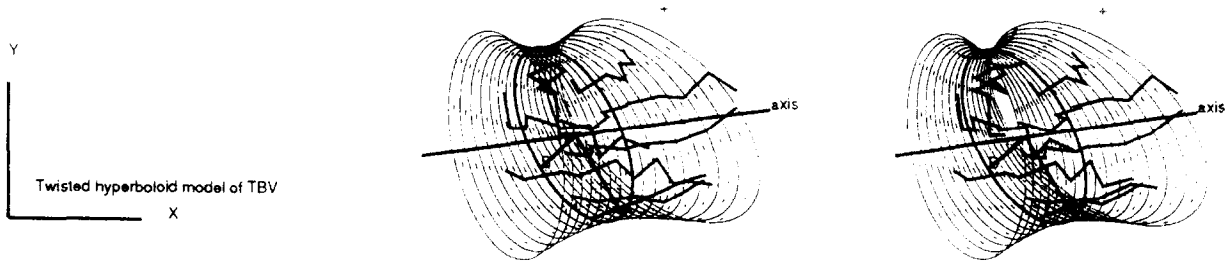

Figure 6. Hyperboloid fitted to the strands of TBV. The principle axis, equatorial plane and major semi-axes are indicated in bold.

topologically equivalent residues, similar contacts between equivalent amino acids in a number of different structures can be elicited by performing a summation, for each cell position in a particular contact map, over all maps. In the resultant plot, conserved contacts between equivalenced residues are readily apparent from large values, relative to the number of structures being compared, in any cell.

Although 25 structures are considered here, each cell in the summed contact plot may not necessarily have a maximum possible value of 25 due to, for example, deletions. In general, the maximum score possible for each (i, j) cell is $[N-(d_i-d_{ji})]$, where N is the total number of sequences in the alignment, d_i the number of deletions in alignment position i, and d_{ji} is the number of deletions in alignment position j that do not occur in position i.

3. Results

(a) *Primary and secondary structures*

The primary sequences of the structures listed in Table 1 were aligned in the eight major strand regions of their respective jellyrolls according to hydrogen bond patterns as described in Methods. The corresponding main-chain superpositions for all the structures over both sheets are shown in Figure 3 and the alignment is given in Figure 7. Boxed residues indicate jellyroll flap regions which are strand segments not participating in the basic wedge composed by the respective BIDG and CHEF sheets. The average length of specific strands over all the structures and associated standard deviations are also given. Similar data are shown for the four wedge loops (W1 to W4, facing the 5-fold axis in viral icosahedra) and the other strand connections (loops joining the C and D, E and F, and G and H strands). The mean hydrophobicity is given below each alignment position. The hydrophobicity values for each of the 20 amino acids as determined by Manavalan & Ponnuswamy (1978) were normalized according to Argos (1987). After averaging the hydrophobicity values at each alignment site, the fraction was rounded to the nearest tenth and then multiplied by 10 and reported as such in Figure 7. All alignment positions with a value of 6 or greater were considered strongly hydrophobic. Strand sites participating in alternate strongly hydrophobic positions are indicated with a (+) where non-hydrophobic sites were considered to have a mean value of 5 or less. Symbols given between aligned residues along a sequence indicate insertions or bulges.

Dayhoff *et al.* (1983) calculated an amino acid exchange matrix containing weights expressing the preferences for the various residue substitutions. These were determined from sequence alignment of 71 different protein families. An evolutionary tree was constructed using the sum of the Dayhoff weights for each aligned residue match over all possible sequence comparisons as shown in Figure 7. The Dayhoff additions for each strand were themselves simply combined for a given jellyroll pair and then divided by the number of matched residue pairs to yield a pairwise similarity matrix. The actual Dayhoff weights utilized are given and recommended by Altschul (1991) and referred to as the PAM120 substitution scores. The similarity matrix over all the jellyroll pairs was then used as input to the computer package BIOPAT of Hogeweg & Hesper (1972). Agglomerative clustering based on complete-linkage (Lance & Williams, 1967) was performed where the distance between the most remote individuals in subgroups was used to construct a tree as illustrated in Figure 8. This approach is amenable for clustering very distantly related objects as it yields smaller subgroups and does not segregate according to closest neighbors. Other clustering techniques including group averaging (Lance & Williams, 1967), the neighbor-joining method (Saitou & Nei, 1987), Ward's (1963) averaging and single linkage (Lance & Williams, 1967) were also employed to generate dendograms. The jellyroll relationships elicited from the various techniques were generally consistent, especially for the picornaviruses, but the complete-linkage approach resulted in a tree most easily interpreted relative to known icosahedral and oligomeric structural associations. The optimal method to use in constructing phylogenetic trees from sequence alignments remains controversial (Felsenstein, 1988).

Dendograms were also made using the r.m.s. distance of superposed C^{α} atoms in each of the two sheets and over all pairs of jellyroll cores as a cluster criterion. Once again the complete-linkage technique produced the most sensible tree which was largely consistent with that obtained from the

```
          N term.   ---------------- B ----------------                W1  ------ C ------                CD
MEV-1       h   40                S P I.A   F                 6        20        N S V I L   5        h  47
RHV-1       h   74                A C.H V   T E I Q N        10        13    L F N D W K I   7        h  13
R1A-1       h   74                G C.H I   S R I K V        10        10    N F T K W K I   7        h  13
PLV-1       h   84                A C.T I   M T V D N        10         9    L F A V W K I   7        h  13
MEV-2  e e      52   S E R+  E R Y Y T F K V N D W T S      20         5    E Y I R I P L   7   h    h  17
RHV-2  e e h    53   S K P+  C R F Y T L D S K T W T T      20         3      G W C W K L   6   h    h  15
R1A-2  e e h    53   T Q P+  N R F Y T L E S K H W N G      20         3      G W W W K L   6   h    h  17
PLV-2  e e h    53   T E P+  C R F Y T L D T V S W T K      20         2    R G W W W K L   7   h    h  15
MEV-3       h   47             A Q:T F I+  Y I E A S N      22         7      L A V Y Q V   6   h    h  17
RHV-3       h   46             I Q:T L I+  L I P L N A      27         4    E Q V.G T N L   8   h    h  17
R1A-3       h   46             C Q:T L I+  T V Q L Y N      28         5    Q K V.S I K V   8   h    h  17
PLV-3       h   46             A E:T M I+  R V R L S D      28         5    D P I.C L S L   8   h    h  18
BMV-A           8                N Q.A T   V R T            8        10  F C Q F S V D L   8   e e  h  27
BMV-B          10             L N R W M G K L T F P Q      12        12  S R.V K R M P L   9   e e  h  22
BMV-C          22             L K.S Q S   K I A L          10         7      T I L.S D L   7   h    h  24
SBV     e e    64  G V T I L S H C E L S T E L A V         16         3  I V V T S E L V   8        h  14
TBV     e e   101  G V T V T S H R E Y L T Q V N N         16         3  F V V N           4        h  29
BBV            92    V T R K D V L N Q   S I N F T A      15         2  D T F I L I A P   8   e e    49
STV     h      24  E H K R F A L I N S.N   T N            14         0    A:G T V Q N L   9        16
AHV-1   h      96  S T Y F D I R G V L D   R G            13         2  F K P Y:T A Y N  10        11
AHV-2         660             Y L S A A.M   L Y P          10         3  N A T N.P I S I   9         4
HMG           115    G T L E F I T E     F T W            12        37  N V T M P N N     7         0
TNF            11      P V A H V V A N P   Q              10   e e  25          Q L V V   4         3
GAP            18                H I H   K Y P            6         2      S T L I H Q   6         4
CNA             2          T I V A V E L   D T            9   e e  37  T A H I I Y N S   8   e e h  30

Hydrophobicity        4 3 5 3 5 6 5 4 5 6 4 3 6 2 7 2 3                4 4 5 5 4 6 3 8
Pattern                             .       .     +   +                            +   +
Core
Ave. length                                                   14.88     9.16                7.16      18.08
                                                               (6.5)    (10.0)               (1.4)     (11.5)
```

```
                  ---------------- D ----------------                W3  --------- E ---------                EF
MEV-1             P F V Y Y K C D L E V T L S P H   16                 5      G L L V R W C P   8         23
RHV-1             L F T Y V R F D S E Y T I L A T   16                12    L V V Q A M Y V    8      h  20
R1A-1             L F T Y V R F D S E I T L V P C   16                 9  H I V M Q Y M Y V    9      h  20
PLV-1             F F T Y S R F D M E L T F V V T   16                13    Q V Y Q I M Y V P  9      h  19
MEV-2             R H Y L V K T G W R V Q V Q C     15                 6  A G S L L V F M A P E 11  h e h  45
RHV-2             F H S L G R S G Y T V H V Q C     15                 6  S G C L L V V V I P E 11  h e h  55
R1A-2                 F L G R S G Y T V H V Q C N   14                 5  Q G T L L V A M I P E 11  h e h  57
PLV-2             Y H Y L G R S G Y T V H V Q C N   16                 5  Q G A L G V F A V P E 11  h e h  64
MEV-3             N F A Q Y R G S L V Y T F V F T   16                 5  K G K F L I A Y T     9      h  16
RHV-3             Y Y T H W S G S L R F S L M Y T   16                 5  S A K L I L A Y T     9      h  16
R1A-3             Y Y T H W T G S L R F S F M F C   16                 5  T L K L L L A Y T P P 11      h  14
PLV-3             Y Y T H W A G S L K F T F L F     15                 7    G K L L V S Y A P    9      h  15
BMV-A             V A A W K K G C L M V K V V M S   16                12      L V Q V F L T      7         10
BMV-B             M W R Y F V G D L V F E V S K M   16                 5  K C T V S F F I A     9         14
BMV-C                         M I K M V A T V       8                 7    I A L A C A M N    8      h  17
SBV               N W S K Y A W V.I R Y T Y L P S   17                 5  S G A I H M G F Q Y  10      h  17
TBV                     Q Y S F.S V V L D Y V P L   14                 5  V G R V A L Y F D K  10      h  17
BBV                         R Y A.M N V G I Y P T   12                 8      S I T V W K C      7         35
STV               Q V R I V S H K L H V R G T A     15                 3  S Q T F R F I W F R D 11    h h  35
AHV-1                 E W E Q T E D S G R A V A     13  Loop - 1   201      L M Y+N M G       11         7
AHV-2                     W A A F R.W A F T R       11  Loop - 3    35  F K K V A I T F D     9         2
HMG               D N F D K L Y I W G I H H         13     h        15  G R.T V S T R         8         0
TNF               G L Y L I Y S Q V L F K G Q       14                 7  L L T H T I S R I A  10         6
GAP                           E T L Y Y I V         7                 2        S V A V L I K  7         8
CNA                   E W V R V G L S A S T         11                 7  T I.S W S F T S K L K 12         9

Hydrophobicity    5 6 3 5 6 3 4 3 7 4 8 3 8 5 5 3                  3 5 4 8 4 8 5 6 6 3 1
Pattern             .     .       +   +   +                            +   +   +   .
Core                                      *                                *
Ave. length                                                   14.16     8.08                9.40      21.64
                                                              (2.5)     (6.4)               (1.4)     (16.9)
```

```
                  ----- F -----                   W4   ---------- G ----------                GH
MEV-1             P Q V Y S         5                 8        I S F V V   +L P A  16         23
RHV-1             P S V F F K V     7                 0    G D.S R F S V   +Y N Q  19         22
R1A-1             M S I F W Q H     7                 3      F P R F S I   +Y Y M  17         22
PLV-1             P S I F Y T Y     7                 3      A P R I S V   +Y S H  17      h  30
MEV-2             P H Q F L N L     7                 3  T T V D L E V P Y+S      16          6
RHV-2             P H Q F I N L     7                 3  N T A T I V I P Y+D S M  18          4
R1A-2             P H Q F I N L     7                 3  N S A T L I V P Y+D S M  18          4
PLV-2             P H Q I I N L     7                 3  N C A T L V L P Y+D S M  18      h   4
MEV-3             T Y A I W D L     7                 3  S S Y S F T V   +R M V  18         11
RHV-3             T H V V W D I     7                 3  S T I V M T I P W+R Y T  18          8
R1A-3             T H V V W D V     7                 3  S T I S L V V P W+R L T  18          8
PLV-3             T H V I W D I     7                 3  S S C T M V V P W+R Q T  18      h   8
BMV-A             C R W T K         5                 5  W E L I F P I E V+E      17         11
BMV-B             P H K L V         5                 6  E K V V L K F     +W S T  18         13
BMV-C             D C E L W N P     7                 3  K A M T M S F              7      h  19
SBV               G Y V T G P V     7               h 21      I T I A L     +Y P P  17      h h 22
TBV               V L K E T A       6                 4        A M L R I              5         22
BBV               D N F S E S       6                 3    G V F S Q S V C           8      e e 48
STV               L.D V T L N C     8                 5  S I K D R I I N L.G Q I  13         16
AHV-1             L N A V V         5   Loop - 1    198  R Y V P F H I Q V P Q      11         12
AHV-2             V S W+E F E I    17   Loop - 4    125          F Y I P E          5         22
HMG                 R S Q Q T I     6                26  R I S I Y W T I V K        9          2
TNF               V N L.S.I K S     9                14  W Y E P I Y L G G.F Q L   13          2
GAP               L.Y L             4                 2      G D F I G              5         11
CNA               L H F M F         5         e e e 38  S V G R A L F V.V H I      13          6

Hydrophobicity    4 3 6 6 6 1 8                       2 4 6 3 7 5 8 3 7 3 3 6
Pattern             . . + +                             +   +   +   +     .
Core                * *                                     *   *
Ave. length                       6.88                7.17                14.08      14.24
                                 (2.3)               (9.0)                (4.7)     (10.4)
```

Fig. 7

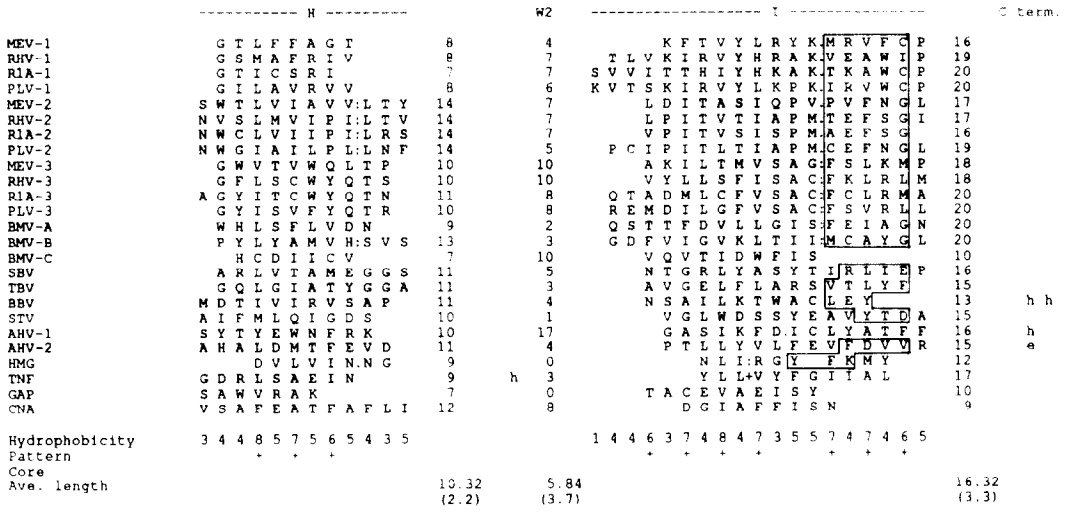

Figure 7. Multiple sequence alignment for the 8 major strands in the 25 jellyroll structures. The lengths of each strand and loop segment in the alignment follows that segment. Major secondary structures (helix, h and strand, e) in the 4 wedge loops (W1 to W4), CD, EF and GH loops and at the structure termini are indicated (mean lengths and associated standard deviations are also given beneath each column of length values). In the alignment insertions and deletions (indels) are indicated by special symbols: indels of length 1 (.); indels of length 2 (:); and (+) longer indels. Under the alignment the mean and relative hydrophobicity of each column is printed. A + symbol is used to indicate the pattern of alternating hydrophobicity along the sequence where the relative hydrophobicity is greater than or equal to 6. Isolated regions of strong hydrophobicity are indicated (.) The 6 core residues discussed in the text as involved in numerous and consistent side-chain contacts are highlighted (*). If loops involved indels more than 100 residues in length, these entries were not included in the calculation of average lengths.

sequence similarity score and led to nearly the same evolutionary scenario for jellyroll development (*vide infra*).

(b) *Residue contacts*

Contact matrices for intra-jellyroll side-chain or main-chain interactions were effected for each of the 25 aligned sequences as explained in Methods. Since alignment positions were structurally equivalenced, the individual contact plots for the various structures were summed. Only matrix cell values greater than 12 and involving inter-strand residue associations were examined as intra-strand contacts were expected and are numerous. If the values in the overall contact plot were summed along each matrix row, six-strand residues made the most consistent inter-strand associations; they are marked as asterisks (*) in Figure 7 alongside the heading marked "core"; the row sums ranged from 87 to 41. Two sites were in each of the strands F and G and one site was in each of the D and E strands. The individual contacting residue pairs with highest sums (values of 17 to 19) included the asterisked sites. Only three other pairs fell into the highest sum category but their specific sites were not represented in the largest row sums. The most significant and most conserved side-chain pairwise contacts are structurally illustrated in Figure 9. It is clear that in strands F and G, the four residues, two from each strand, interdigit in the wedge core while in the D and E strands the two side-chains contact directly across from each other.

Statistics were also gathered as to the preferred residue–residue pairs by amino acid type. The three largest were Val-Leu, Val-Ile, and Leu-Leu with respective observation counts of 33, 29 and 19. It is clear that the large hydrophobic residues such as Phe or Trp are not in the majority.

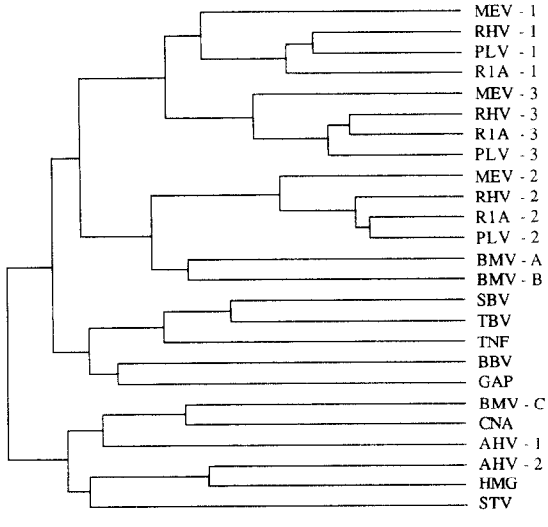

Figure 8. Relative evolutionary relationships between the 25 jellyroll structures derived from a pairwise structure similarity matrix (Dayhoff *et al.*, 1983; Altschul, 1991) calculated from the aligned sequences of Fig. 7 using the complete linkage clustering algorithm (Lance & Williams, 1967) as found in the BIOPAT computer program package (Hogeweg & Hesper, 1972, 1984).

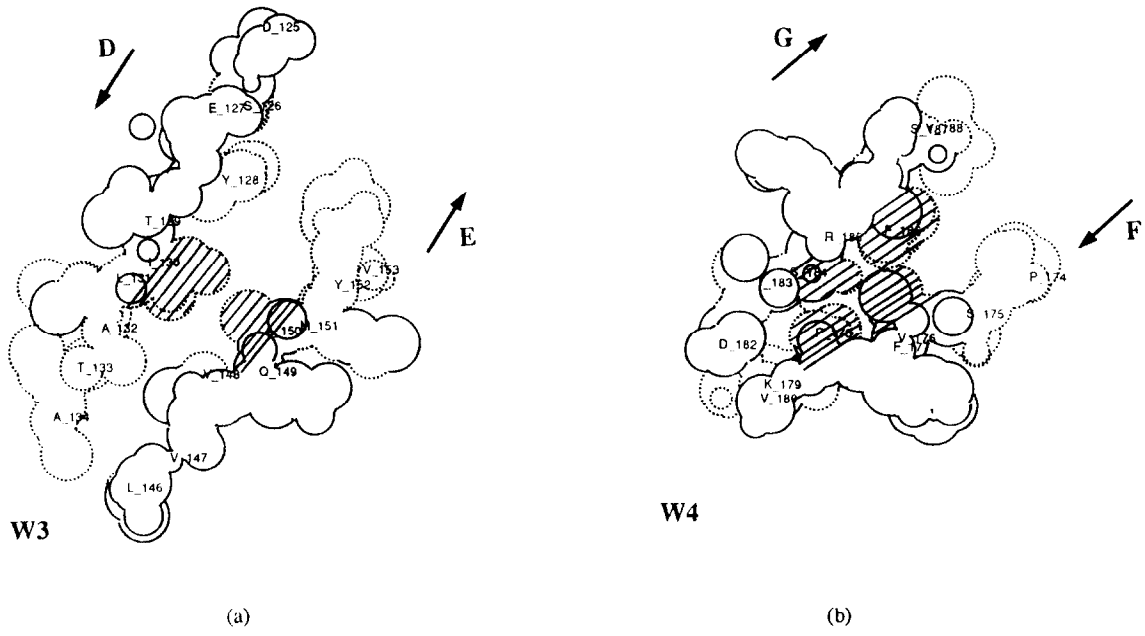

Figure 9. Contour plots of the 6 core residues in RHV-1. (a) A contour plot of the residues in strands D and E of the jellyroll showing the packing between residues I130 and S150, which in the text are designated with site numbers 40 and 49. The relative position of the wedge loop W3 is indicated. (b) A contour plot of the residues in strands F and G, at the base of the wedge. Extremely tight packing is evident between residues V176, F178, S184 and F186, which are designated in the text with alignment position numbers 57, 59, 64 and 66, respectively. The wedge effect is also pronounced as the distance between adjacent residues on the opposite strands increases in moving away from the wedge loop W4 in the direction of the G strand.

(c) *Intra-sheet–inter-strand distance*

Estimates of the smallest separation distance between strands (inter-strand) in each (intra-sheet) of the BIDG and CHEF sheets are listed in Table 2. The values show considerable overall consistence with an average of 4·2 Å, clearly expected as a result of the inter-strand hydrogen bonding. Other than this, no general pattern emerges from the values listed. There is, however, a rough trend for the VP2 subunits of the picornaviruses to show slightly higher distances than are seen for the VP1 and VP3 subunits. The results also show that the outer-sheet strand pairs have greater separation than the inner or middle strands. For example, the B-I and D-G (outer strands) have respective separations of 4·37 and 4·42 Å while the inner I-D pair shows only 3·6 Å. The results are similar for the CHEF sheet. The stronger hydrophobic core contacts as well as the solvent and RNA exposed faces of the wedge may cause the stretching.

(d) *Intra-sheet–inter-strand angle*

Table 3 gives an account of the measured twist angles between successive pairs of strands down each sheet. The angles are measured in an anticlockwise direction consistent with the sheets left-handed twist (Richardson, 1981) when viewed perpendicular to the sheet plane and strand directions. The average twist angle for both sheets is approximately 19°. All of the measured values lie in the range of about 0 to 57°, although 80% of them can be confined to the range 10 to 30°. It is noteworthy that the average twist decreases in moving from the solvent exposed strands to the buried RNA-interface strands.

In terms of the total twist of the sheets (the sum of the successive twists down the sheet), the BIDG sheets show a somewhat stronger twist. Although the range of twist angles is large, there is a clear tendency for a decrease by about 5° between successive angles while going down the sheet. The VP1 subunits are exceptional in that they show a consistent decrease and then increase in twist angle, albeit by less than 10° in all cases. The picorna VP3 subunits all show an exceptionally large twist angle of about 52° between the B and I strands.

In the CHEF sheet, the picorna VP3 subunits tend to show a smaller twist angle between strands pairs CH and EF than between H and E. This pattern is also followed by BMV-C, TBV and BBV amongst the virus structures and GAP from the non-virus structures. In the remainder of the structures the twist angle between the C and H strands is larger than the angle for the HE or EF pairs.

(e) *Inter-sheet distance*

Table 4 lists the values measured for the approximate inter-sheet distance (D). This value is extremely well conserved with an average value of 11 Å and a standard deviation of less than 1 Å. AHV-1 shows the largest value of about 13 Å and TNF the smallest value of about 9·1 Å, which are

Table 2
Intra-sheet–inter-strand distance

File	B–I	I–D	D–G	C–H	H–E	E–F
MEV-1	4·249	3·449	3·842	4·791	4·137	4·808
RHV-1	3·621	2·932	4·308	4·382	3·590	3·697
R1A-1	4·349	2·984	3·813	4·416	3·862	4·272
PLV-1	3·664	2·924	4·508	4·278	4·000	4·113
MEV-2	4·094	3·988	4·679	4·514	4·136	4·436
RHV-2	4·033	3·934	4·594	4·682	4·028	4·493
R1A-2	4·075	3·928	4·760	5·065	3·918	4·418
PLV-2	4·174	3·845	4·592	4·908	3·905	4·534
MEV-3	6·272	2·953	4·028	4·101	3·921	4·438
RHV-3	5·771	3·365	3·824	4·363	3·696	4·470
R1A-3	6·113	3·160	3·966	4·213	3·510	4·580
PLV-3	5·670	3·301	3·989	4·077	3·723	4·411
BMV-A	5·541	2·565	4·436	5·237	3·687	4·544
BMV-B	4·343	3·208	4·311	4·644	3·552	5·194
BMV-C	3·794	3·354	4·567	4·358	4·050	4·733
SBV	2·741	4·011	5·192	4·087	4·026	3·821
TBV	3·278	4·240	5·819	4·614	3·724	4·848
BBV	3·849	4·087	5·524	4·015	4·201	3·996
STV	4·288	4·597	4·635	4·334	4·468	4·556
AHV-1	4·452	4·075	4·520	3·949	2·991	6·469
AHV-2	3·305	3·976	4·075	4·774	4·705	3·900
HMG	4·677	4·868	3·127	4·297	4·642	4·163
TNF	4·233	3·926	4·388	4·708	4·406	4·650
GAP	4·582	2·870	4·866	4·131	4·088	4·447
CNA	4·197	5·087	4·132	4·536	4·859	4·789
Average	4·37	3·67	4·42	4·46	3·99	4·51
Standard deviation	0·87	0·64	0·56	0·33	0·41	0·52

Table 3
Intra-sheet–inter-strand angle

File	B–I	I–D	D–G	C–H	H–E	E–F
MEV-1	31·978	20·693	15·265	27·806	18·873	13·162
RHV-1	26·349	16·299	19·988	24·910	24·196	25·976
R1A-1	21·191	17·601	21·566	25·999	17·946	19·690
PLV-1	26·263	17·386	23·587	25·508	19·528	20·972
MEV-2	30·396	15·669	10·712	21·235	11·762	15·726
RHV-2	25·992	15·162	9·871	17·673	11·965	14·159
R1A-2	25·352	15·378	8·897	20·995	11·800	13·440
PLV-2	26·137	15·438	9·065	17·956	11·472	13·168
MEV-3	57·246	24·259	21·414	15·384	18·106	16·708
RHV-3	55·280	23·152	21·997	13·838	19·576	15·774
R1A-3	49·708	23·432	20·342	20·585	20·018	15·570
PLV-3	49·427	22·891	21·315	18·237	18·418	15·612
BMV-A	18·053	18·658	16·254	23·464	16·867	13·079
BMV-B	22·845	25·240	18·553	23·577	19·938	20·801
BMV-C	20·028	24·047	16·462	0·696	22·214	13·794
SBV	25·436	15·854	14·143	26·795	17·149	13·184
TBV	22·608	17·603	15·824	13·816	14·463	8·073
BBV	21·982	12·315	11·677	14·548	20·163	16·538
STV	2·146	13·061	9·498	27·442	14·929	10·030
AHV-1	40·616	28·433	25·576	25·598	4·039	12·907
AHV-2	22·371	20·821	12·905	14·841	11·560	5·800
HMG	24·553	23·339	21·589	15·802	7·740	7·156
TNF	9·487	8·152	6·359	29·025	14·624	1·921
GAP	35·651	38·419	22·717	3·620	32·893	17·998
CNA	15·518	4·230	7·066	12·426	2·032	8·527
Average	28·26	19·10	16·11	19·27	16·09	13·99
Standard deviation	13·12	6·71	5·67	7·11	6·30	5·13

Table 4
Inter-sheet orientation and distance

File	A1	A2	A3	D
MEV-1	−134·821	−59·073	12·964	11·302
RHV-1	−156·882	−38·606	42·743	9·285
RIA-1	−150·728	−46·809	35·176	9·861
PLV-1	−144·529	−38·012	49·038	9·591
MEV-2	−163·957	−32·879	24·113	10·690
RHV-2	−161·698	−29·066	24·436	10·755
RIA-2	−163·423	−28·227	22·923	10·810
PLV-2	−162·855	−29·202	24·593	10·945
MEV-3	−151·513	−45·239	12·356	11·549
RHV-3	−148·310	−40·401	25·963	11·810
RIA-3	−150·509	−37·750	28·794	11·554
PLV-3	−148·828	−40·524	26·497	11·591
BMV-A	−164·980	−36·475	5·514	10·958
BMV-B	−163·577	−33·889	0·266	11·595
BMV-C	−131·172	−48·607	24·058	11·758
SBV	−151·558	−12·834	47·960	10·771
TBV	−171·257	−2·439	54·179	12·305
BBV	−165·277	−14·045	24·849	10·839
STV	−169·330	−19·836	15·646	10·394
AHV-1	1·469	−19·445	61·942	13·395
AHV-2	−129·466	−57·203	9·425	10·999
HMG	−179·605	10·589	17·484	10·956
TNF	−170·211	−36·865	5·315	9·121
GAP	−129·449	−15·301	1·761	11·068
CNA	−167·128	−51·559	8·261	11·046
Average	−156·50	−32·15	24·25	11·00
Standard deviation	14·38	16·39	16·30	0·91

Table 5
Inter-sheet–inter-strand distance

File	B–C	I–H	D–E	G–F
MEV-1	9·648	14·092	11·484	10·159
RHV-1	9·133	11·910	10·985	8·566
RIA-1	9·535	12·632	11·324	10·458
PLV-1	8·878	12·319	11·370	9·913
MEV-2	11·085	14·626	12·153	8·338
RHV-2	11·168	14·495	12·099	8·193
RIA-2	10·794	14·644	12·056	8·197
PLV-2	11·263	14·662	12·343	8·050
MEV-3	13·396	14·269	11·649	8·589
RHV-3	15·048	14·602	11·303	8·304
RIA-3	14·893	14·301	11·197	7·821
PLV-3	14·848	14·398	11·219	7·886
BMV-A	10·019	12·467	11·859	10·068
BMV-B	9·874	14·265	12·013	10·688
BMV-C	12·934	14·378	11·724	8·090
SBV	11·567	13·430	12·195	9·483
TBV	16·211	12·871	12·143	7·931
BBV	10·916	12·515	12·462	11·457
STV	11·660	10·917	10·215	8·558
AHV-1	8·504	11·828	17·882	24·348
AHV-2	12·206	11·001	13·862	12·469
HMG	15·624	9·987	12·018	10·783
TNF	11·994	13·274	8·408	9·032
GAP	13·510	11·997	13·190	9·171
CNA	17·031	13·853	15·179	17·481
Average	12·07	13·19	12·09	10·16
Standard deviation	2·39	1·34	1·67	3·53

amongst the largest and smallest jellyrolls, respectively.

(f) *Inter-sheet orientation*

Table 4 also lists values for the horizontal wedging (A1), rotation (A2), and vertical wedging (A3), as defined under Methods, between the planes representing the BIDG and CHEF sheets. There is a rough overall agreement in the relative organization of the planes. The orientations correspond to a horizontal wedging with the point of the wedge at the side of the jellyroll containing the short BC, HI, DE and GF loops (W1 to W4, respectively); a slight vertical wedging with the sharpest part of the wedge occurring between the G and F strands at the base of the jellyroll; and an anticlockwise rotation of the CHEF plane with respect to the BIDG plane. In spite of the general agreement in the spatial location of the sheets, explicit values for the measured angles show great diversity except when taken in the context of the genetically defined subclasses. The most uniform class is that of the picorna VP2s. For these structures, the sheets have a rotation angle of about −30°. They tend to be inclined towards each other in a horizontal sense with an angle of about 17°, and in a vertical sense with an angle of about 23°. The VP3 subunits have a similar vertical wedging of about 25° but a much larger horizontal wedging of about 35°. The rotation angle is about −40°. In the VP1s, the horizontal wedging has measured values of roughly 30°, rotation angles of about −37° (with the exception of MEV-1) but quite diverse vertical wedging. There is an approximate linear correlation between the magnitude of the horizontal wedging and rotation angle between the BIDG and CHEF sheets. Generally, the larger the rotation, the larger the horizontal wedging.

(g) *Inter-sheet–inter-strand distance*

Estimates of the distances between all combinations of strands from opposite sheets indicates a relationship between the pairs of strands B/C, I/H, D/E and G/F, these pairs tending to be closer. This is somewhat intuitive given that these related strands occur in equivalent positions within the context of their own sheet. Table 5 shows the values measured for the inter-sheet–inter-strand distances for these pairs. There is a general trend for the separation distance to first increase and then decrease while going down the sheet from the solvent exposed face to the RNA abutting side. This is suggestive of a bulging between the two sheets, in a cross-sectional view, peaking between strands I and H. The strand distance between pairs I/H and D/E, which form the middle of the wedge, show the highest overall consistence amongst the viral capsids (AHV-1 excluded) with respective average (and standard deviation) values of 13·44 Å (1·22) and 11·78 Å (0·72). The consistence for the G/F distance is also greatly enhanced by only consider-

Table 6
Inter-sheet–inter-strand orientation

File	A. Horizontal wedging				B. Rotation			
	B–C	I–H	D–E	G–F	B–C	I–H	D–E	G–F
MEV-1	29·714	0·229	2·452	19·038	−6·158	−32·883	−24·281	−5·889
RHV-1	13·020	0·000	2·116	19·523	−2·113	−34·303	−35·812	−6·726
R1A-1	14·976	0·000	0·000	18·590	−8·396	−38·552	−34·068	−0·179
PLV-1	20·172	0·000	1·242	14·790	−3·653	−38·885	−36·851	−1·919
MEV-2	11·264	0·000	0·000	3·151	−9·982	−23·199	−12·245	−0·852
RHV-2	14·445	0·000	3·845	7·949	−6·947	−22·208	−11·270	−2·626
R1A-2	9·780	0·000	2·782	7·356	−4·959	−20·748	−10·504	−1·738
PLV-2	13·865	0·000	2·578	6·305	−6·737	−21·946	−10·663	−1·440
MEV-3	47·426	0·000	0·000	1·933	−56·334	−37·179	−17·415	−0·282
RHV-3	44·142	0·000	0·160	1·670	−57·801	−39·316	−18·866	−0·148
R1A-3	36·047	0·000	0·000	3·328	−43·814	−40·063	−18·696	−0·184
PLV-3	39·309	0·000	0·000	0·588	−46·162	−39·872	−19·779	−1·200
BMV-A	6·990	7·335	2·552	8·292	−9·691	−17·833	−20·523	−0·826
BMV-B	8·747	0·000	1·475	5·626	−12·245	−30·487	−18·235	−4·293
BMV-C	19·497	0·000	2·404	2·350	−25·276	−40·672	−16·048	−0·217
SBV	7·439	4·978	10·389	8·393	−8·039	−15·052	−6·570	−18·081
TBV	15·786	4·243	4·912	1·268	−21·106	−15·581	−7·654	−8·501
BBV	12·509	6·199	9·047	19·426	−6·303	1·313	−9·372	−7·082
STV	17·633	2·029	5·402	6·891	−12·096	−14·225	−12·189	−7·645
AHV-1	0·000	0·000	21·280	121·023	−48·422	−41·409	−57·043	−82·211
AHV-2	25·004	0·000	0·000	0·000	−26·440	−38·022	−45·383	−21·447
HMG	19·827	17·015	8·812	−3·527	−55·002	28·082	9·401	−7·318
TNF	24·093	3·245	0·000	0·000	−14·288	−25·916	−18·504	−16·159
GAP	74·141	40·231	39·754	23·891	−7·365	−5·318	−15·379	−52·291
CNA	20·810	24·678	23·458	37·492	−20·693	−22·348	−28·685	−30·579
Average	21·87	4·41	5·79	13·41	−20·80	−25·17	−19·87	−11·19
Standard deviation	15·75	9·34	9·21	23·86	18·36	15·76	13·53	18·65

ing these structures yielding an average value of 9·16 Å (1·3). In contrast the B/C distance still maintains its high diversity with an average value of 11·75 Å (2·1) explainable by virtue of these strands playing host to interactions with the diverse surface loops of the protomers. The picorna VP1 and BMV-A subunits exhibit less of a bulging effect than the other subunits which make up their capsids, since the I/H distance in these cases is about 2 Å less. Of the $T = 3$ viruses, both BBV and SBV show the bulging while in STV and TBV this is not the case as a consequence of the B/C distance being the largest. Here, only a continual decrease in the strand pair distance, while going down the sheet, is observed.

(h) *Inter-sheet–inter-strand orientation*

The horizontal wedging angle, A4, of the strand pairs BC, IH, DE and GF is given in Table 6A. The most striking feature is the fact that the strands I and H generally show none, or very little, horizontal wedging between them; i.e. the strands lie in essentially parallel planes. A similar situation is seen for the D and E strands. The I/H and D/E strand pairs form the interior of the jellyroll; apparently side-chain packing and consequently structural stability are affected by the possibility of interacting strands. Table 6B, lists the rotation angle, A5, between the same strand pairs as above. There is a clear trend for anticlockwise rotation of the C, H, E and F strands with respect to the B, I, D and G strands, respectively.

(i) *Kernel volume*

Estimates of the volume enclosed by the wedged packing of the BIDG sheet to the CHEF sheet are given in Table 7; the flap region has been excluded. The Table shows a wide fluctuation in total volume, ranging from 3354 Å^3 for GAP to 5844 Å^3 for STV. The mean value of the volume is 4946 Å^3 with a standard deviation of about 600 Å^3. The larger molecular weight structures do not necessarily produce the largest jellyrolls. The picorna VP1s and VP3s are consistently smaller than the VP2 subunits, which average 5483 Å^3.

(j) *Hyperboloid characteristics*

The mean orientation of each of the strands used in the hyperboloid fit with respect to the hyperboloid axis L is given by the angles W1 and W2 in Table 8. The average inclination of each strand either towards or away from L is given by W1 while W2 describes the rotation of the strands with respect to L. The values for W1 show a good overall consistence with the average inclination being small at about 3°. STV shows the smallest value for W1 as expected due to the flat nature of its sheets. The W2

Table 7
BIDG CHEF sheet volume

File	B	C	D	E	F	G	H	I	Total
MEV-1	437	141	728	611	369	512	624	1191	4613
RHV-1	598	540	776	435	473	567	748	1244	5381
R1A-1	812	359	855	576	575	500	642	963	5282
PLV-1	789	328	896	574	500	400	656	913	5056
MEV-2	919	503	465	544	457	449	971	937	5245
RHV-2	778	707	668	515	458	427	1004	931	5488
R1A-2	933	707	668	544	458	399	868	905	5482
PLV-2	748	707	668	531	458	425	1193	958	5688
MEV-3	168	461	708	640	617	406	631	759	4390
RHV-3	168	828	839	628	668	338	675	821	4965
R1A-3	168	683	875	704	641	335	676	716	4798
PLV-3	168	669	708	577	668	287	700	797	4574
BMV-A	659	511	937	449	525	736	748	936	5501
BMV-B	975	470	682	800	312	511	777	987	5514
BMV-C	625	493	740	622	516	543	426	786	4751
SBV	543	277	574	673	273	336	682	614	3972
TBV	996	545	511	577	434	258	721	731	4773
BBV	1116	458	674	480	482	878	697	606	5391
STV	864	349	632	804	763	708	788	936	5844
AHV-1	595	459	316	591	276	476	780	629	4122
AHV-2	1062	600	800	512	542	534	805	633	5488
HMG	358	742	969	672	630	802	500	453	5126
TNF	367	167	1047	502	596	859	426	745	4709
GAP	370	601	371	282	433	358	323	616	3354
CNA	308	168	355	874	573	459	966	437	4140
Average	620·96	498·92	698·48	588·68	507·88	500·12	721·08	809·76	4945·88
Standard deviation	297·61	185·32	186·53	123·51	121·42	169·59	187·89	196·31	600·43

rotations fairly consistently center around $-16°$. The negative tilt corresponds with a right-tilted antiparallel barrel (Chou *et al.*, 1990).

The magnitudes of the major and minor hyperboloid semi-axes correspond to the dimensions of the equatorial ellipse. Measured values of these parameters (A and B) are given in Table 8. Both sets of values show a general agreement over all structures. The major axis has an average length of roughly 10 Å while the minor axis length averages near 6 Å.

Table 8 also shows values for the area (Ar) of the equatorial ellipse. TBV has the smallest area,

Table 8
Twisted hyperboloid parameters

File	W1	W2	A	B	Ar	R	E	T
MEV-1	4·558	−18·051	10·382	6·256	204·046	0·603	0·798	3·361
RHV-1	6·449	−19·753	9·823	5·762	177·806	0·587	0·810	3·188
R1A-1	4·727	−20·176	10·175	6·729	215·094	0·661	0·750	2·382
PLV-1	7·216	−20·467	9·790	5·925	182·235	0·605	0·796	2·399
MEV-2	1·258	−15·444	12·188	6·046	231·498	0·496	0·868	3·933
RHV-2	2·609	−13·427	10·708	5·843	196·568	0·546	0·838	3·715
R1A-2	1·775	−13·442	10·912	6·104	209·232	0·559	0·829	3·960
PLV-2	2·132	−13·271	10·498	6·077	200·427	0·579	0·815	3·516
MEV-3	1·061	−13·019	8·880	6·759	188·537	0·761	0·649	−0·107
RHV-3	0·768	−14·379	9·232	6·758	196·012	0·732	0·681	9·458
R1A-3	1·419	−16·971	9·039	6·637	188·454	0·734	0·679	9·008
PLV-3	1·754	−16·555	9·141	6·754	193·977	0·739	0·674	9·141
BMV-A	2·103	−15·334	10·916	6·996	239·915	0·641	0·768	3·414
MBV-B	3·123	−20·357	13·707	5·927	255·234	0·432	0·902	3·063
BMV-C	3·453	−18·214	9·775	6·596	202·549	0·675	0·738	2·560
SBV	3·426	−12·370	10·162	5·915	188·850	0·582	0·813	1·222
TBV	3·576	−13·325	10·052	5·588	176·456	0·556	0·831	1·716
BBV	4·382	−9·635	9·219	6·758	195·735	0·733	0·680	4·050
STV	0·363	−12·667	8·337	7·460	195·388	0·895	0·446	0·743
AHV-1	−7·532	−28·780	14·401	7·440	336·610	0·517	0·856	1·632
AHV-2	3·471	−22·139	9·631	8·180	247·478	0·849	0·528	3·898
Average	2·48	−16·56	10·33	6·50	210·58	0·64	0·75	3·63
Standard deviation	2·84	4·25	1·47	0·64	35·45	0·11	0·11	2·53

around 7 Å^2, and AHV-1 the largest, nearly 337 Å^2, providing a range of values over 160 Å^2. Despite this large overall range, more than two-thirds of the values can be confined to about a 40 Å^2 range from 176 Å^2 to 216 Å^2. The structures that do not fall in this range are BMV-A, BMV-B, MEV-2, AHV-1 and AHV-2. Generally then, the jellyroll structures do maintain a reasonably similar cross-sectional area as has been previously pointed out by Richardson (1981).

Values for the eccentricity (E) of the equatorial ellipse were calculated and shown in Table 8 with the mean near 0·75. The picorna VP2 subunits are seen to have the highest eccentricity.

All but one of the twist values (T) in Table 8 are small and right-handed. The exception for MEV VP3, though negative, is very close to zero. The handedness of the twist matches the inherent right handed twist of β-sheets in the direction of the strands. Most of the structures show a twist of about 3°/Å, corresponding to a helical pitch of about 120 Å.

4. Discussion

(a) *Primary structure*

The primary sequences of the structures in Figure 7 show considerable diversity. This is consistent with the notion that three-dimensional biological structures are generally conserved on a much longer time scale than their constituent amino acid sequences (Rossmann & Argos, 1981). Thus, very distantly related proteins can share the same topology but differ greatly in sequence. The sequence patterns in Figure 7 should be useful in recognizing jellyroll folds in newly determined sequences. The alignments generated here show general agreement with previous structural alignments performed for the picornaviruses RHV and MEV and the plant viruses TBV, SBV and STV by Rossmann and co-workers (Krishnaswamy & Rossmann, 1990; Luo *et al.*, 1987; Arnold & Rossmann, 1990; Rossmann *et al.*, 1983); however, exceptions are found at four insertion sites. Also the alignment involving the BIDG sheet of STV requires shifts of two to four residues relative to the earlier matches. The bulge residue 143 in strand B of MEV-1 is assigned differently here although the bounding residues still have the same alignment as reported earlier. A similar situation is found for RHV-3 and R1A-3 in the C strand and between SBV and STV in the F strand.

An alignment derived from a rigid-body superposition of foot-and-mouth disease virus (FMDV) onto MEV and RHV has also been reported (Acharya *et al.*, 1989). In this alignment, the structural equivalencing yielded a shift of one residue in strands C and H for the VP1 and strand C for the VP2 subunits of MEV and RHV relative to the alignment given here. Such a shift puts part of the CHEF sheet out of register so that residues pointing into the core of the jellyroll in one structure are equivalenced with residues pointing to the exterior of the other protein. Utilizing equivalences derived from separate hydrogen bond patterns, as was done here, circumvents this potential hazard. Further, from the values listed in Table 4, it is evident that the precise orientations of the BIDG sheets with respect to the CHEF sheets vary greatly, much more so than the values for strand orientations within a sheet (Tables 2 and 3). Thus, it seems more natural to equate the less diverse, or perhaps more constrained, sheet topologies separately than to attempt a single rigid-body superposition of the whole structure.

A pure rigid body superposition technique has also been applied to equivalence TNF with other jellyrolls (Jones *et al.*, 1989). Although it was found there that equivalences could be derived for over 60% of the TNF residues (with r.m.s. deviations frequently less than 3·5 Å), no alignment was given and so the effectiveness of these superpositions could not be judged. It is possible that the derived equivalences for TNF suffered from similar inconsistencies as those in FMDV. Consequently, separate sheet superposition were performed here.

The sequence alignment of Figure 7 shows several distinct and conserved alternating patterns of hydrophobic/hydrophilic residues along adjacent alignment columns (hydrophobic sites annotated with a plus (+) in Fig. 7). These patterns were recognized by designating match columns with mean hydrophobicity values of 6 or greater as hydrophobic and those with an average of 5 or less as hydrophilic where the amino acid hydrophobicities of Manavalan & Ponnuswamy (1978) have been normalized as by Argos (1987). The residue extent of the patterns grows in size when proceeding from the solvent-exposed strands B and C (top of barrel, see Fig. 1B) to those facing the nucleic acid (bottom of barrel) as exemplified by strands F and G. The pattern is most extensive in strand I. The hydrophobic positions (the + sites in Fig. 7) participating in the zig-zag scheme are never fully populated by hydrophobic residues (AIVLMCWFHY) as 10, 5 and 4% of the amino acids at these sites are comprised by the groups (QNST), (KRDE) and (PG), respectively. Alternating hydrophobic sites with polar exceptions being largely Gln, Asn, Ser or Thr should be salient markers for jellyroll strands. The hydrophilic positions which are sandwiched between the hydrophobic sites or form the N and C termini of the strands display different compositional trends where 41% of the residues are nonetheless hydrophobic, 29% and 22% come from the (QNST) and (KDER) groups, respectively, and only 8% are Pro or Gly. Such patterns can also point strongly to strand regions and should be useful in structure prediction. Proline, glycine and asparagine residues are also often found adjacent to the strand N and C termini.

(b) *Flaps*

As previously noted, the jellyrolls are best envisioned as a wedge formed by two intersecting sheet

planes with a sometimes addition of a flap region, resulting in a triangular sheet arrangement encompassing a common core. When the flap is missing, the strand loops (EF, CD and GH) on the non-wedged side of the wedge enshroud and complete the core. The flap segments are boxed in Figure 7. It is clear that the hydrogen bond pattern in a given strand is broken at its flap edge where a strong bend occurs, after which the strands continue to yield the flap extended structures.

The bend involves bulges in the I and G strands as well as a "corner" residue which is the first residue after the flap in the D strand. The character of the residues in these regions has a strong preference for turn type residues: nearly 50% of them occurring in the set (SNDGP) considered as having a high turn preference (Palau *et al.*, 1982) in comparison to an expected 29% distribution (McCaldon & Argos, 1988). A bulge is not a prerequisite for a bend in a strand as is seen for the D strand in Figure 7, which generally maintains a characteristic in-out strand pattern. Bend deviation angles (as measured by the angle between vectors fitted to the I, D and G strands and their respective flap region counterparts) of between 35 to 85° are permissible in such cases. Small bulges of one or two residues can also result in this range of bend. Angles greater than 90°, however, require larger bulges of about eight residues and can involve a single left-handed helical loop as seen in the G strand of the picornaviruses where the main-chain loops back over itself before continuing as part of a strand. Bends in strands can occur off to the left, as in the I and G strands, or off to the right as for the D strand. It is interesting to note that the right-handed bend can achieve large deviation angles of almost 90° without a bulge while left-handed, non-bulged, bends attain angles of only about 50°.

The flap segments, if they occur, are always confined to the BIDG sheet; however, they do not display a convincing hydrophobic/hydrophilic pattern. In principle, the flap extensions could emanate from one side of the CHEF sheet. It will be interesting if any such structures are found. The CHEF sheet of TNF does contain an extension in this vicinity but rather than sealing off the core of the wedge, the extension twists away from BIDG more radically emphasizing the wedge effect. Consequently this region was not considered a flap. Three or four flap spans occur in each of the VP structures as well as in SBV, TBV, BBV and STV. Their first evolutionary appearance would be then expected during the development of a $T = 1$ icosahedral particle (*vide infra*).

(c) *Loops*

In Figure 7 the strand-connecting loops have been designated W1 to W4 and CD, EF and GH. The latter are named according to the strands they connect while the former are at the wedge (W) intersection of the jellyroll and join extended structures, as illustrated in Figure 1A. The average loop lengths and accompanying standard deviations amongst the superposed structures are given in Figure 7. Of the structures studied, it is clear that the wedge turns, averaging about seven residues in length, are the shortest in all the viruses except AHV-1 and AHV-2. These wedge loops pack around icosahedral 5-fold axes (Liljas, 1986; Rossmann & Johnson, 1989) and cannot grow too large without disrupting the symmetric protein shell unless the insertions bend away from the axis and are accommodated by the rest of the structure. The recently solved structures of canine parvovirus (CPV: Tsao *et al.*, 1991) and the bacteriophage ΦX174 (FIX: McKenna *et al.*, 1992) have wedge loops with long insertions. Although both form icosahedrally symmetric shells, W1 of CPV contains two short strands which veer away from the wedge to form a surface feature with the EF and extensive GH loops. Likewise, W2 of FIX F protein involves 110 residues which bend up and back over the jellyroll, again forming a complex surface feature. W3 of CPV is unique in that its β-ribbon insertion bends upwards and follows parallel to the 5-fold axis resulting in a cylindrical structure about the axis in the complete capsid. Only W4 at the base of the wedge and at the capsid interior seems less amenable to allow insertions.

The strict icosahedral symmetry axes, especially the 2-fold, are approximately orthogonal to the strand directions for all the viruses but AHV where the extended conformations are more radial; i.e. they parallel the symmetry axes. In the latter case, the symmetrical wedging constraints do not exist and the loop lengths are accordingly larger or more variant as in the case of the non-wedge loops in the jellyroll structures. The same is true for simian virus 40 (Liddington *et al.*, 1991), which also exhibits a radial strand arrangement. The secondary structure assignments of the seed storage protein phaseolin (Lawrence *et al.*, 1990), with strands running orthogonal to a pseudo-diad axis, shows that neither the α or β domain jellyrolls have large wedge loops. The CD, EF and GH connectors are largest (mean around 18 amino acids) for the viruses with EF generally displaying the greatest lengths with an average of 21. About 80% of these loops contain only α-helices with the remainder consisting of strands or both strands and helices.

In the viruses, all extensive loop excursions occur at the surface of the capsid where packing constraints are less restricted. Even when the loop initiates at a subunit interface or close to the capsid inner surface, the loop directs itself towards the relative freedom of the surface. This has the added advantage of allowing the virus to avoid immune surveillance and dictate host cell specificity *via* mutations in these loops while still preserving its essential capsid core.

β-Bulges (interrupting loop excursions within strands) are found most often in strands B and C. Nonetheless, isolated bulges are found in almost all the other strands, albeit the hairpin E and F extended structures displaying the fewest inter-

ruptions. Such observations should be useful in engineering the viral jellyrolls as they point to the most constrained or relaxed structural regions.

(d) *Barrel shapes*

It has been shown by Novotny *et al.* (1984) that β-barrel proteins can be modeled as twisted hyperboloids (strophoids). Lasters *et al.* (1988) have computed hyperbolic parameters for several such parallel-stranded β-barrel proteins. A similar approach was used here to derive the related parameters in jellyroll structures. Compared with the parallel β-barrel proteins, the jellyroll structures generally show a consistently higher eccentricity as a consequence of a longer major semi-axis (Table 8). For the same reason, the area of the equatorial ellipse is also larger (Table 8). The average angle between the strands and the hyperboloid axis (the barrel tilt) is also markedly different. For typical barrel proteins an angle of about $-35°$ is seen (Chou *et al.*, 1990; Lasters *et al.*, 1988) but in the case of the jellyrolls the angle tends to be much smaller lying roughly between -13 and $-19°$.

Lesk *et al.* (1989) have shown that in α/β-barrel proteins the packing of the barrel interior includes 12 side-chains somewhat segregated into three layers. Each layer contains four side-chains donated by every second strand of the barrel. They propose two classes of packing. In the first class, which contains barrels like rubisco and triosephosphate isomerase, the central layer is formed by residues from the first, third, fifth and seventh strands with the upper and lower layers using contributions from the second, fourth, sixth and eighth strands. In the second class, the roles played by the alternating strands are reversed so that the central layer is formed by residues from the even strands. No such layering was found for jellyroll structures as is evident from contact maps which show that no single position in the B-strand is in contact with residues from the H, F and D strands simultaneously in more than one structure. The same holds true for the C, E, G and I strands. Clearly then, jellyrolls are not comparable with parallel-stranded β-barrels.

In a study of the relative orientation of close packed β-sheets, Chothia & Janin (1981) modeled their packing as a twisted prism with a rectangular cross-section that rotates about the twist axis of the sheets. They found from their sample that sheets tended to pack with an angle of about $-30°$. At the interface of the sheets, they described the arrangement of the side-chains as pointing directly towards each other with rows of side-chains being approximately aligned. Cohen *et al.* (1981) examined the face-to-face packing of two primarily antiparallel β-sheets in a number of structures by modeling each sheet as a plane. They reported a standard packing geometry where the sheets were separated by between 8·3 to 10·3 Å and were rotated with respect to one another anticlockwise by 20 to 50°. The sheet interface was described as involving the interdigitation of large and small residues in different sheets. The results of both of these studies are generally in agreement with the results obtained here. Visual inspection of the packing of the side-chains in the interior of the jellyroll showed the interdigitation model as favored (e.g. Fig. 9). Nonetheless, the jellyroll structure is seen as a wedge shaped polyhedron of two sheets as illustrated in Figure 1B.

(e) *Specific side-chain contacts and packing*

Contact maps between inter-sheet and inter-strand side-chains were effected (see Methods). The maps were constructed by marking each alignment position along the column and row borders of a matrix and then placing within the matrix cells a value of 1 if contact is made between the two specific side-chains, each residing in an opposite sheet. Such matrices were then added for each jellyroll structure, resulting in an overall contact plot emphasizing the packing consistency amongst the various folds studied.

The most interactions involved residues in opposite strands but on the same level (see Fig. 9): i.e. between strands G and F, D and E, H and I, and B and C. The side-chains making the most consistent contacts with others have been marked with an asterisk (*) in Figure 7 and include two positions in strand G with the more N-terminal residue (numbered as 64) making a total of 47 contacts and the more C-terminal (numbered as 66) showing 87 total contacts. Similarly, the more C-terminal asterisked site in strand F (numbered 59) and two single sites in D and E (numbered 40 and 49, respectively) displayed respective interaction counts of 54, 42 and 41. For the overall contact plot the most consistently associating residue pairs included 64–57, 64–59, 66–57 and 40–49 (where numbered site 57 corresponds to the more N-terminal asterisked site in strand F of Fig. 7) scoring 17 or larger out of a maximum possible of 25. The 57 and 59 sites in strand F and 64 and 66 positions from strand G participate in alternating and interdigitating side-chain packing as illustrated in Figure 9(b). Residues numbered 40 and 49 simply contact each other across opposite strands D and E in opposing sheets (Fig. 9(a)). All these positions are also hydrophobic members of the conserved zig-zag strand patterns and thus are likely to provide the most stabilizing and consistent hydrophobic side-chain interactions in the jellyroll folds. Any substitutions must be carefully considered. The preferred contacting amino acid pairs over all the jellyroll inter-sheet inter-strand residues were Val-Leu, Val-Ile and Leu-Leu. The absence of large aromatic amino acids is surprising as they would provide the largest interacting surfaces or faces for pairwise residue association.

Antiviral compounds have been shown to bind to a pocket in VP1 of entero and rhino viruses (Smith *et al.*, 1986; Badger *et al.*, 1988). Although the VP2 and VP3 subunits can be superimposed onto VP1, the former do not bind the compounds (Arnold &

	MEV-1	RHV-1	R1A-1	PLV-1	MEV-2	RHV-2	R1A-2	PLV-2	MEV-3	RHV-3	R1A-3	PLV-3	BMV-A	BMV-B	BMV-C	SBV	TBV	BBV	STV	AHV-1	AHV-2	HMG	TNF	GAP	CNA	
MEV-1	5.66	0.41	0.13	0.73	-0.49	-0.70	-0.71	-1.14	-0.71	-0.85	-0.55	-0.34	-0.99	-0.88	-1.59	-0.54	-1.03	-0.75	-1.22	-1.92	-0.95	-1.41	-0.80	-1.78	-1.97	MEV-1
RHV-1	1.06	5.55	2.79	2.82	-0.97	-0.67	-0.68	-1.17	-0.22	-0.88	-0.76	-0.83	-1.29	-1.46	-1.39	-0.76	-1.06	-0.85	-1.89	-1.93	-0.94	-1.69	-1.32	-1.04	-1.43	RHV-1
R1A-1	0.94	1.00	6.01	2.25	-1.40	-1.06	-1.23	-1.43	-0.89	-0.81	-0.94	-1.02	-1.58	-1.64	-1.29	-1.16	-1.20	-1.15	-1.61	-1.70	-1.33	-2.09	-1.32	-1.12	-1.16	R1A-1
PLV-1	1.05	0.65	0.82	5.73	-1.08	-0.24	-0.65	-0.96	0.09	-0.62	-0.42	-0.36	-1.21	-1.25	-1.35	-0.56	-0.58	-0.74	-1.68	-2.28	-1.14	-2.12	-1.09	-1.59	-1.20	PLV-1
MEV-2	1.38	1.50	1.19	1.21	5.49	1.99	2.43	2.11	-1.15	-0.78	-0.83	-0.49	-0.91	-0.76	-1.21	-1.76	-1.53	-0.71	-1.20	-1.60	-1.34	-1.80	-1.55	-1.49	-1.88	MEV-2
RHV-2	1.47	1.60	1.27	1.26	0.64	5.56	3.89	3.63	-1.07	-1.01	-1.12	-0.55	-1.13	-1.10	-1.38	-1.85	-1.40	-1.41	-0.99	-1.81	-1.60	-1.77	-1.76	-1.18	-1.43	RHV-2
R1A-2	1.35	1.51	1.22	1.18	0.72	0.42	5.63	3.96	-1.10	-1.25	-1.23	-0.79	-0.80	-1.15	-1.12	-1.87	-1.39	-1.10	-1.25	-1.88	-1.70	-1.94	-1.64	-1.32	-1.68	R1A-2
PLV-2	1.50	1.69	1.41	1.45	0.71	0.41	0.51	5.79	-1.22	-1.23	-1.24	-0.81	-0.85	-1.04	-1.27	-1.96	-1.17	-1.23	-1.68	-1.66	-1.63	-1.92	-1.86	-1.41	-1.62	PLV-2
MEV-3	1.57	1.59	1.25	1.19	1.36	1.44	1.38	1.56	5.41	1.34	1.67	1.52	-1.19	-1.33	-1.29	-0.43	-0.50	-1.16	-1.75	-2.24	-0.88	-1.89	-1.54	-0.62	-1.65	MEV-3
RHV-3	1.41	1.47	1.35	1.14	1.40	1.47	1.32	1.52	0.59	5.55	3.49	3.06	-1.13	-0.92	-0.77	-0.82	-0.90	-1.42	-1.58	-1.55	-0.59	-1.67	-2.03	-1.66	-1.46	RHV-3
R1A-3	1.25	1.43	1.40	1.05	1.50	1.50	1.35	1.51	0.60	0.50	5.70	3.26	-1.16	-1.21	-1.05	-0.81	-1.09	-1.53	-1.64	-1.57	-0.75	-1.91	-1.61	-1.43	-1.62	R1A-3
PLV-3	1.31	1.50	1.42	1.14	1.28	1.36	1.24	1.31	0.64	0.43	0.46	5.58	-1.05	-0.74	-0.79	-0.98	-0.88	-1.25	-1.76	-1.32	-0.61	-1.88	-1.67	-1.46	-1.81	PLV-3
BMV-A	1.82	1.90	1.76	1.88	1.99	2.12	1.57	2.23	2.36	2.13	2.02	1.92	5.48	-0.79	-0.98	-1.37	-0.73	-1.32	-1.76	-1.07	-1.03	-1.16	-1.25	-1.58	-1.72	BMV-A
BMV-B	1.51	1.34	1.36	1.19	1.66	1.76	1.76	1.93	1.59	1.55	1.55	1.74	2.16	5.69	-0.90	-1.20	-0.38	-1.33	-1.53	-1.84	-1.43	-1.31	-1.67	-1.36	-1.33	BMV-B
BMV-C	1.17	1.54	1.20	1.19	1.29	1.11	1.37	1.36	1.28	1.17	1.08	1.19	1.65	1.42	5.50	-1.07	-1.50	-1.28	-1.63	-1.52	-1.34	-1.58	-1.62	-2.00	-0.55	BMV-C
SBV	1.40	1.51	1.49	1.41	1.37	1.43	1.40	1.53	1.58	1.51	1.42	1.41	2.14	1.79	1.57	5.27	0.50	-1.10	-1.18	-1.90	-0.77	-0.83	-0.70	-1.22	-2.10	SBV
TBV	1.34	1.62	1.44	1.32	1.41	1.38	1.38	1.64	1.85	1.76	1.65	1.69	1.89	1.90	1.41	0.97	5.10	-1.19	-1.90	-1.87	-0.64	-0.91	-0.89	-1.54	-1.48	TBV
BBV	1.33	1.80	1.36	1.50	1.80	1.76	1.74	1.76	2.09	2.04	1.90	1.78	1.93	1.68	1.88	1.58	1.58	5.18	-1.62	-1.56	-1.71	-1.28	-1.26	-1.24	-1.09	BBV
STV	2.24	2.29	2.07	2.19	1.80	1.74	1.71	1.89	2.06	2.50	1.98	2.17	1.46	2.22	1.75	1.86	1.73	1.72	5.30	-1.82	-1.38	-1.61	-1.14	-1.35	-1.12	STV
AHV-1	3.41	3.26	2.53	2.66	3.82	3.20	3.03	3.88	2.55	4.10	2.07	3.98	3.93	3.94	2.40	2.97	2.92	2.83	2.64	5.52	-1.76	-1.55	-1.51	-1.73	-1.08	AHV-1
AHV-2	1.59	1.87	1.73	1.68	2.50	1.53	2.46	2.40	1.99	1.86	2.00	1.73	1.45	1.86	1.74	2.57	2.26	1.74	2.39	1.48	5.54	-0.17	-0.27	-1.18	-0.84	AHV-2
HMG	1.75	2.07	2.10	2.12	2.04	1.94	1.97	2.27	1.82	2.33	2.03	2.15	1.83	2.06	1.58	2.38	2.53	2.40	2.06	2.14	2.59	5.53	-1.55	-1.50	-1.88	HMG
TNF	2.38	2.32	2.03	2.36	2.33	2.24	2.36	2.33	2.18	2.67	2.12	2.61	2.11	2.30	2.16	2.60	1.66	1.93	1.95	2.74	2.17	2.33	5.33	-1.12	-1.26	TNF
GAP	1.44	1.48	1.39	1.40	1.37	1.52	1.60	1.46	1.55	1.61	1.64	1.60	1.96	1.43	1.66	1.81	1.85	2.08	1.45	2.02	1.52	1.55	2.12	5.33	-1.35	GAP
CNA	2.40	2.58	2.49	2.60	3.03	2.52	2.98	3.16	2.50	2.63	2.79	2.51	2.74	2.59	1.93	3.27	3.10	3.34	4.20	2.38	2.02	2.43	3.59	2.30	5.12	CNA
	MEV-1	RHV-1	R1A-1	PLV-1	MEV-2	RHV-2	R1A-2	PLV-2	MEV-3	RHV-3	R1A-3	PLV-3	BMV-A	BMV-B	BMV-C	SBV	TBV	BBV	STV	AHV-1	AHV-2	HMG	TNF	GAP	CNA	

Figure 10. The scoring matrix used to derive phylogenetic trees for the jellyroll structures. The upper right triangle of the matrix, including the diagonal, shows the normalized sum of Dayhoff PAM120 mutation weights (Altschul, 1991) over the strand alignments given in Fig. 7 for all pairs of jellyrolls. The lower left triangle, excluding the diagonal values, shows the C^{α} atom r.m.s. distances averaged over the superpositions for the CHEF and BIDG sheets.

Rossmann, 1990). An examination of the internal volumes of the subunits shows that all contain internal cavities. Thus, no explanation for the preference of VP1 for compound binding is offered here other than that the vertical wedging between the two sheets (Table 4) is notably larger for these subunits, allowing potential ligands greater access to the jellyroll core from the exterior of the virus capsid.

(f) *Evolution of jellyrolls*

The maintenance of a protein's three-dimensional fold has been shown to be a greater requirement during evolutionary time than the conservation of primary sequence alone (Rossmann & Argos, 1981). The main-chain preservation in very distant structures is almost always confined to those segments which contribute side-chains to the globular protein core. The core backbone spans are usually secondary structural units such as helices and strands and rarely display loop conformations where insertions and deletions are prevalent. In the present work, the sequences of the highly conserved and constrained core β-strands have been carefully aligned according to tertiary structural hydrogen bonding patterns and should bear a strong memory of the structural relationships amongst the proteins studied here as well as their development from an ancestral jellyroll. In principle, this evolutionary record can be tapped by examining the often distant sequence relationships elicited from tertiary structural superposition or by comparing various physical and geometrical characteristics of the jellyroll wedges as shape, size or orientation. The ancestral relationships implied by both approaches are subsequently described and contrasted.

A symmetric score matrix was determined for all pairwise sequence comparisons over the structures examined here (Fig. 10). Each matrix cell contained the normalized sum of the Dayhoff mutation weights (Dayhoff *et al.*, 1983; Altschul, 1991) over all the strand alignments shown in Figure 7 for a given pair of jellyroll structures. The complete jellyroll comparison matrix was then submitted to a tree building routine (Hogeweg & Hesper, 1984) using a complete linkage clustering algorithm (Lance & Williams, 1967). The resulting relationships are illustrated in the dendogram of Figure 8 in which there are four major subclusters coming from the first three splits of the tree. The structures in the two subclusters of the upper half of Figure 8 (above and including BMV-B) are designated group I members while the proteins in the remaining two subclusters belong to group II.

It is clear that the capsid proteins appearing in $T = 3$ virus particles utilizing three polypeptide chains with different primary structures (VP1, VP2, VP3) are clustered according to their VP designations and not according to their hosts or symptomatic expression. BMV-A and BMV-B corresponding to VP1 and VP3 (Chen *et al.*, 1989), respectively, by their position in the $T = 3$ icosahedral shell are also members of this major group (group I) consisting of $T = 3$ particles with different chains (herewith designated $P = 3$). Furthermore, BMV-A and -B are part of the VP2 subcluster and may better retain memory of their ancestral fold claimed here to be VP2 (*vide infra*). These segregations would suggest that VP1, VP2 and VP3 most likely diverged from each other before they diverged further to yield different genera. The structural requirements for a stable $P = 3$ particle must first be met during development and only then can specialized alterations for given hosts and cellular receptors follow. This is consistent with the idea put forward by Argos *et al.* (1980) that both the plant and picornaviruses evolved from a common ancestor such as an ancient receptor binding protein which folded with a jellyroll topology. These consistent results would also support the quality of the strand superpositions and alignments presented here (Figs 3 and 7, respectively).

The constituents in the second major group (group II) comprising the two subclusters in the lower half (below BMV-B) of the tree in Figure 8 display various quaternary structures. SBV and TBV, which are paired in the tree, utilize the same polypeptide chain and configure a $T = 3$ icosahedral protein shell. BBV, also a $T = 3$ type, belongs to the same subcluster as SBV and TBV. STV is the only representative of a same-chain $T = 1$ particle and is quite distant from the $T = 3$ cluster but still part of group II. An anomaly is the BMV-C placement which icosahedrally corresponds to VP2 and would be expected to be in group I. The remaining members of group II are all isolated chains that participate in some n-fold symmetric arrangement: TNF, AHV-1 and 2, and HMG form trimers while GAP and concanavalin A are confined to dimers. It is notable that AHV-1, AHV-2 and HMG are in the bottom subcluster. TNF seems misplaced as it has been positioned in the $T = 3$ group. It is also noteworthy that these n-fold chains have their strand directions running approximately parallel to their respective symmetry axes: i.e. the β-structures are radial. On the other hand, all the virus structures (excepting AHV and HMG) display approximately orthogonal strands in their jellyrolls; i.e. icosahedral symmetry axes are approximately perpendicular to the strand directions. The n-fold GAP strands lie between radial and orthogonal.

An examination of the oligomeric interfacing regions in the various jellyroll architectures also supports the group I/group II segregation. Three terms will be used to designate specific outer segments of the jellyroll fold: face indicates the approximately planar and exterior surfaces of the BIDG or CHEF sheets; loop refers collectively to the oligopeptide connectors at the wedge end of the two associating sheets or to the chain segments joining strands at the other wedge end; and edge designates one of the two opposing wedge sides formed by either the B and C or the G and F extended structure pairs. The regions of the wedge shape referred to by these terms are illustrated in

Figure 1B. In the group II folds with strand directions orthogonal to the n-fold symmetry axes, the oligomeric interfaces are either edge/edge (limited to the dimers) or edge/face (confined to the trimers). In the group I proteins with radial strands and which form icosahedra, the association is very different to the group II interactions: namely, face/face for the icosahedral 3- and 5-folds and loop/loop for the 2-folds and quasi-symmetries. The inter-sheet strands in the face/face interacting sheets maintain approximate parallelism.

Palmenberg has deduced a phylogenetic relationship amongst picornaviruses according to the complete VP1 nucleotide alignments (Palmenberg, 1989). The aformentioned results are in agreement with the dendogram presented here in that MEV is clearly more distantly related than are R1A, RHV and PLV. Contra to the Palmenberg tree and to the classification by virus genera, the two VP1 rhinoviruses are not positioned together but are separated by PLV. This discrepancy arises from the fact that only the residues constituting the core of the jellyroll are considered here while Palmenberg considers the entire nucleotide sequence. Further, Palmenberg classifies RHV as the most distant serotype of the rhinoviruses, at about 54% nucleotide sequence identity, while the distinction between the enterovirus and rhinovirus genera is made at the 50% sequence identity level. The Palmenberg picornavirus sequence alignment upon which the phylogeny was based and the broader alignment of Pascarella & Argos (1992) strongly support the conservation of the hydrophobic pattern depicted in Figure 7.

The above observations would suggest the following scenario for jellyroll evolutionary development, albeit with some exceptions, but nonetheless a strong overall consistency. A single jellyroll polypeptide first appeared and acted as an ancester. The exposed strand-connecting loops were likely to be largely composed of hydrophilic residues, typical of monomeric structures. Interaction surfaces were next developed, yielding the n-fold structures as the dimeric concanavalin A, a particularly ancient protein amongst those considered here. Since the loops faced the solvent in the early development, the only oligomeric interfacing possible was edge/edge or face/face, further necessitating the orthogonality of intra-sheet strand direction and symmetry axis. The jellyroll's ability to take on quaternary structure allowed the development of the $T = 1$ shells with their requirements for the same chain to participate in several different symmetry types. The loops would also have to become more hydrophobic to accommodate their interaction at the icosahedral symmetry axes, especially for the 2-fold which is a direct loop/loop interface. Following the establishment of the $T = 1$ particle, the $T = 3$ shell could then come into being where the same chain can also take on quasi-symmetric interactions. A final feat involved the development of the $P = 3$ particles, utilizing regular and quasi-symmetries as well as similarly folded but covalently different chains. The $P = 3$ capsid's appearance would have been a major structural leap and would have required considerable evolutionary time. The major segregation of the group I $T = 3$ structures in the tree of Figure 8 would support this possibility. Though the tree is not entirely consistent with this evolutionary scenario, the overall agreement is compelling.

The picornaviral VP2, as juxtaposed to VP1 and VP3, can be regarded closer to the ancestral fold. There are many features that are unique to VP2 and distinguish it from VP1 and VP3. The VP2 N termini contain strands and helices while VP1 and VP3 display only helical structures. Furthermore, SBV and TBV which show N-terminal strands in common with VP2 are part of the group II structures. VP2s are the chains found along exact 2-fold symmetry axes in the $P = 3$ particles; such dimers are expected to appear early in the jellyroll's development. The group II structures include dimers. Johnson & Argos (1985) have reviewed experiments showing dimers of chains used in $P = 3$, $T = 3$ and $T = 1$ particles to be among the most stable symmetric arrangements. VP2s show consistently larger eccentricity than VP1s or VP3s (Table 8). They also have the largest EF loops (Fig. 7) which could reflect the longer evolutionary existence of VP2. VP1 and VP3 show smaller wedge volumes than VP2, an observation correlated with VP2s greater intra-sheet inter-strand distances. Furthermore, BMV-C, which is a VP2 icosahedral analog, is placed in the group II category containing the oligomeric jellyrolls. BMV-A and -B are also part of the VP2 subcluster and may retain a better memory of their ancestor. Finally, the tree of Figure 8 places the VP2 chains in one subcluster while the VP1s and VP3s are joined together in the other subcluster of the group I proteins.

Trees were also constructed from the C^{α} atom r.m.s. distances, averaged over the superpositions in each of the BIDG and CHEF sheets, used as a similarity criterion (Fig. 10). The resulting dendogram yielded a similar evolutionary scenario as obtained from that based on the sequence alignment information. It is noteworthy that in both cases, an examination of only the protein cores determined by consistent structural equivalences yielded interpretable results. It is thus suggested that the conserved inner protein cores may provide the best evolutionary memory and that sequence similarity and r.m.s. distances from superposed main-chains are good criteria to elicit protein relationships. Using only the protein cores also provides a normalizing mechanism required for cases where radically different lengths of sequence and structural equivalences can occur over various protein pairs. Furthermore, the cores do not include the highly mutable exposed loops.

Attempts were made to build trees from the other geometric and physical jellyroll characteristics calculated in this work (Tables 2 to 8). For each parameter, the values were also sorted according to decreasing size. The sorted list was then investigated for clusters of VP1, VP2 and VP3 chains as

well as the TBV/SBV pairing. In no case were all the VPs segregated into their individual groups, a result that does not even allow a consideration of the adjacency of the three groups in the sorted list as found in the tree based upon sequence alignments. A score of 1 was given if two VPs with the same number designation appeared as an adjacent pair in the descending list. Similarly, a score of 3 was given to a characteristic parameter for three adjacent VPs of the same type. Finally, a value of 4 was awarded for four consecutive same-numbered VPs in the list. The only three characters to pair TBV and SBV and to show a total score of 9 or 10 where the maximum is 12 were the inter-sheet B/C inter-strand distance, the vertical wedging angle for strands D and E and the intra-sheet B/I inter-strand distance. Thus, many physical and geometric characteristics involving the overall size, shape and orientation of protein core structural units are not the best sources of evolutionary memory.

5. Conclusions

A careful consideration of hydrogen bonding schemes, including relaxation of constraints in defining such bonds, provides a convenient mechanism to equivalence structurally proteins with similar main-chain topology as exemplified by the jellyroll architectures studied in this work. Such superpositions have allowed the recognition of the major sheet/sheet protein core, residues essential for its stability, its wedge rather than barrel shape, the possibility of an extended flap region and the calculation and comparison of several core characteristics over the various proteins. It is clear that great variance is possible with evolutionary time. Furthermore, conserved sequence patterns made salient by the structural equivalences are important markers for the recognition of jellyroll folds in newly discovered sequences for sensible engineering of known jellyroll proteins. It is also noteworthy that core sequences and structures aligned from three-dimensional information apparently provide the best memory of evolutionary development despite the potential of several other overall physical and geometric core characteristics. An evolutionary scenario has been given that may represent the movement from the ancestral fold to the architecture of complex modern viruses.

The authors wish to extend thanks to David Thomas for many insightful suggestions. The authors are also grateful for financial support from the German Bundesministerium für Forschung und Technologie in the form of a research grant (FG5-1075). The University of Western Australia has provided partial support for G.C. in the form of a postgraduate research award as has the Deutsche Akademische Austauschdienst (DAAD). Atomic co-ordinates for the tertiary structures of BMV and BBV as well as AHV were kindly and respectively supplied by Jack Johnson at Purdue University and Roger Burnett at the Wistar Research Institute. Thanks also go to Nelly van der Jagt-González and Martina Radtke for assistance in preparation of the manuscript and Figures.

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Edited by F. Cohen

Note added in proof. It should be noted that the native fold of the CNA molecule is a result of post-translational transposition and ligation of an initial polypeptide (Carrington, D. M., Auffret, A. & Hanke, D. E. (1985). Polypeptide ligation occurs during post-translational modification of concanavalin A. *Nature (London)*, **313**, 64–67. However, in this work, we consider only the final molecular structure for comparative and evolutionary purposes.